The Lives of Magnetic Massive Stars

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Queen's University

RASC Kingston Centre
October 15, 2026



ESO/L. Calçada

Who am I?

Observational Stellar Astrophysicist



Transiting Exoplanet Survey Satellite (TESS)

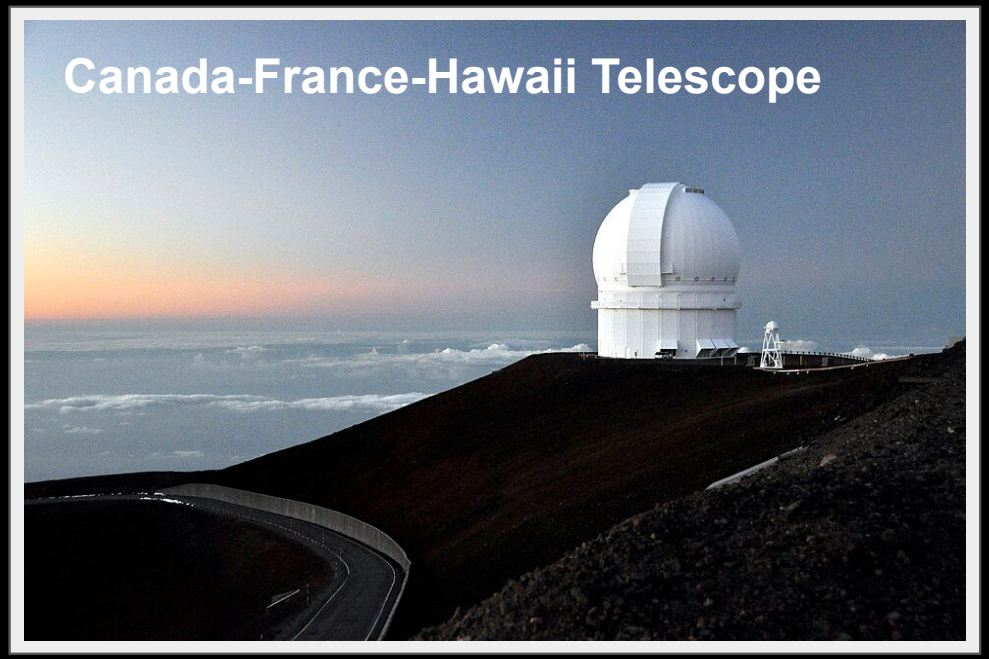
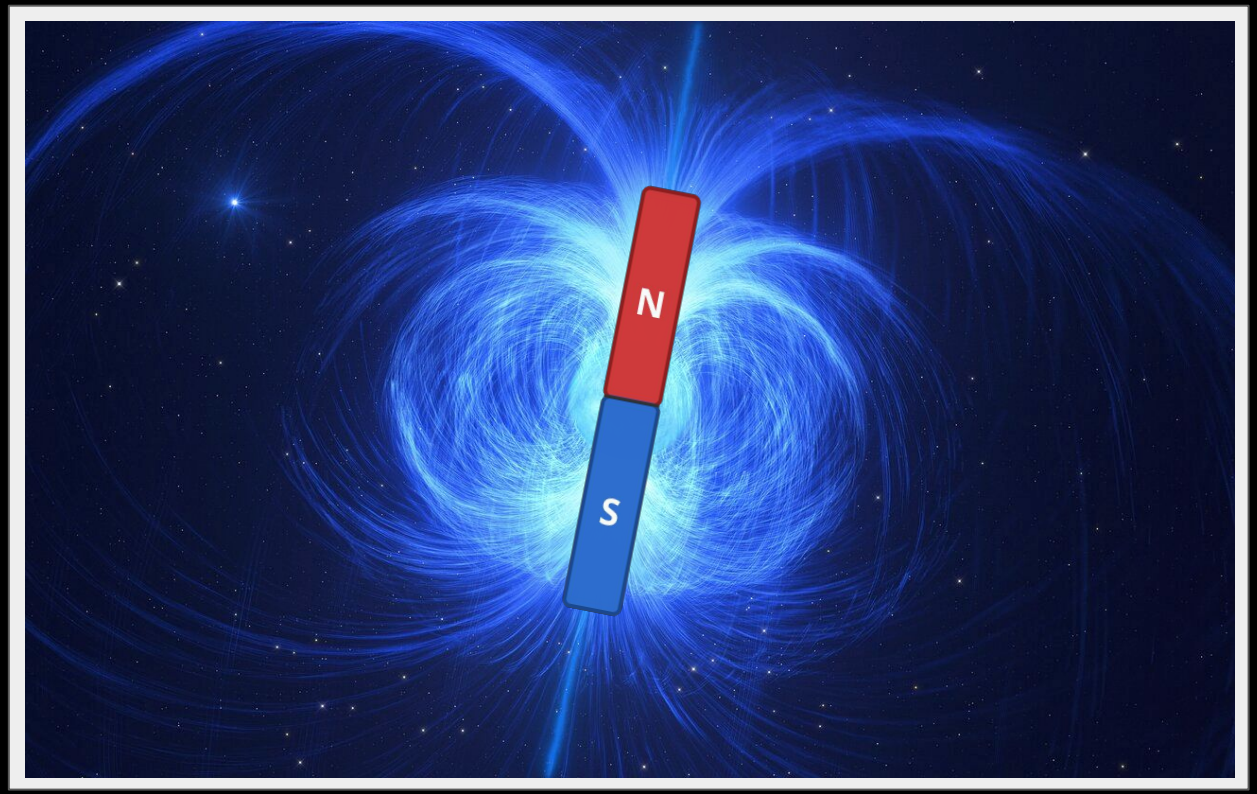


Image Credit: Gordon W Myers

What do I study?

Magnetic Massive Stars



ESO/L. Calçada

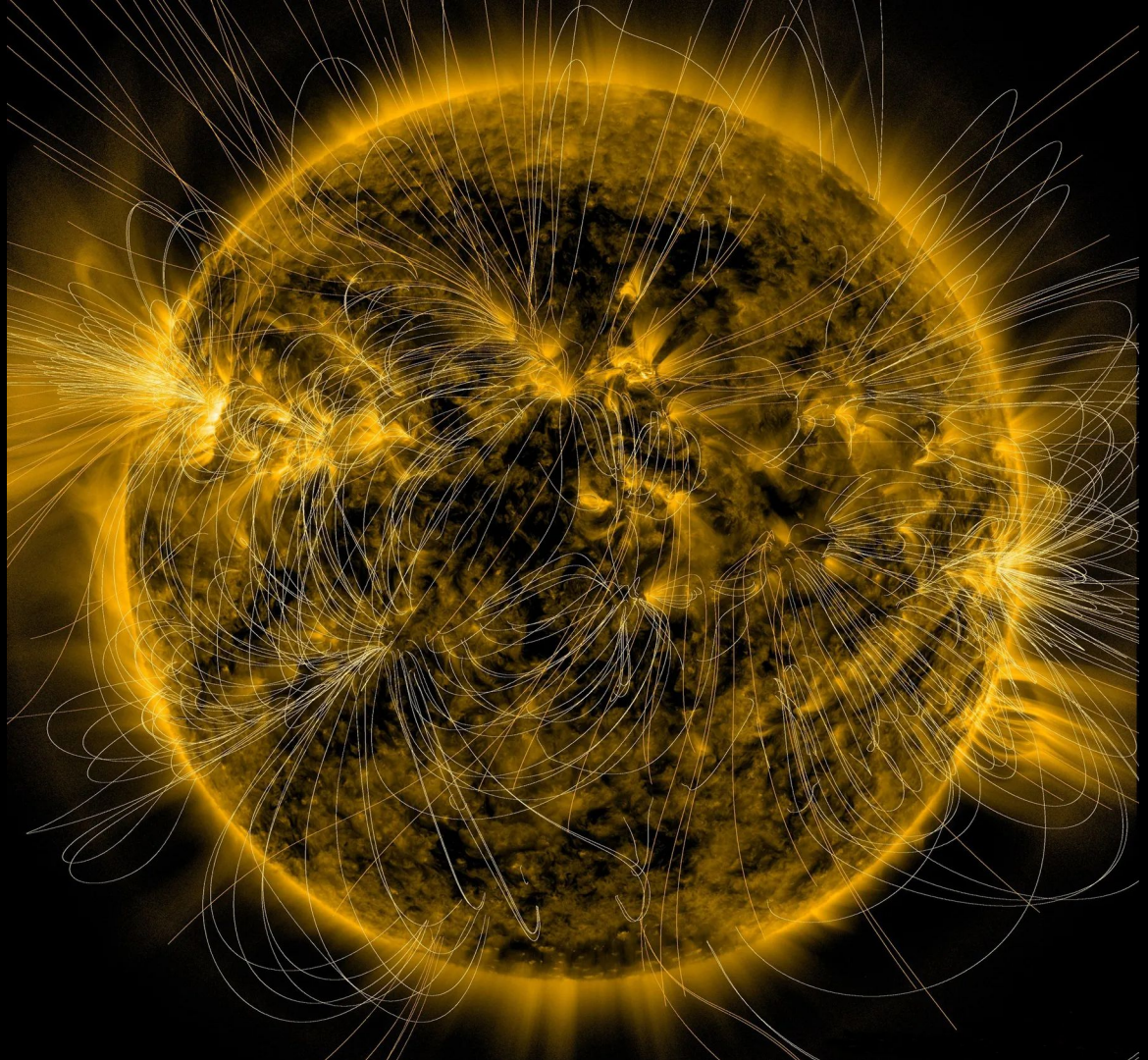
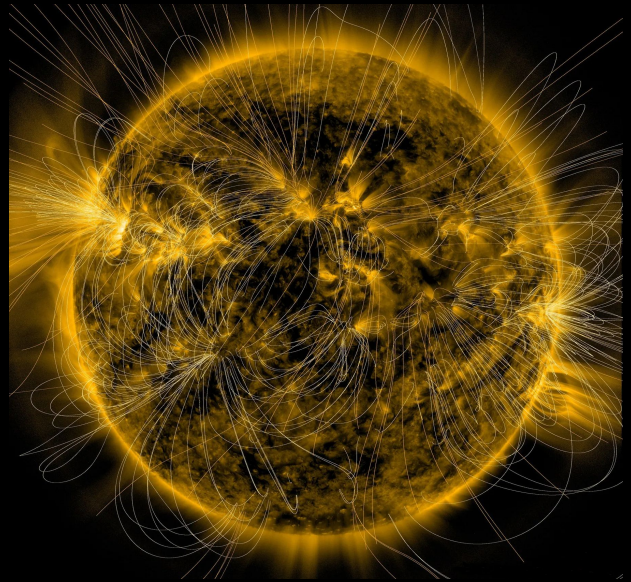
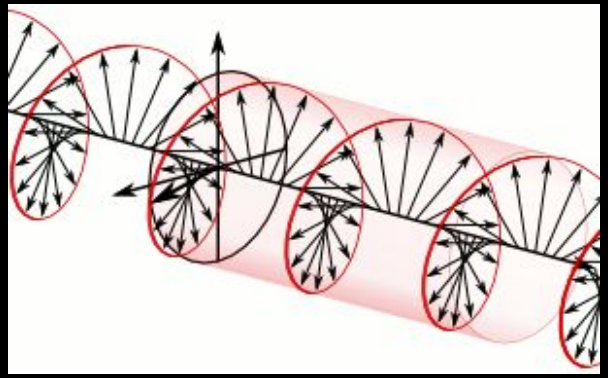


Image Credit: NASA/SDO/AIA/LMSAL

How do we measure stellar magnetic fields?



Polarized Light





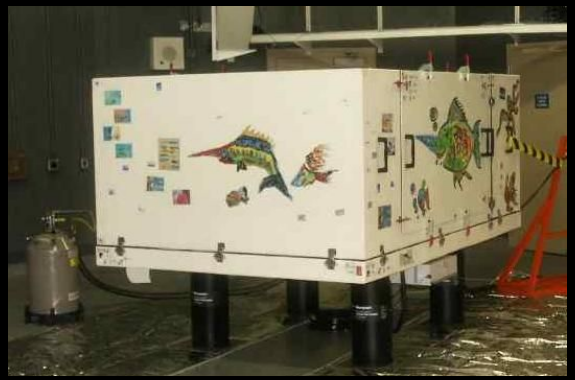


Image Credit: CFHT

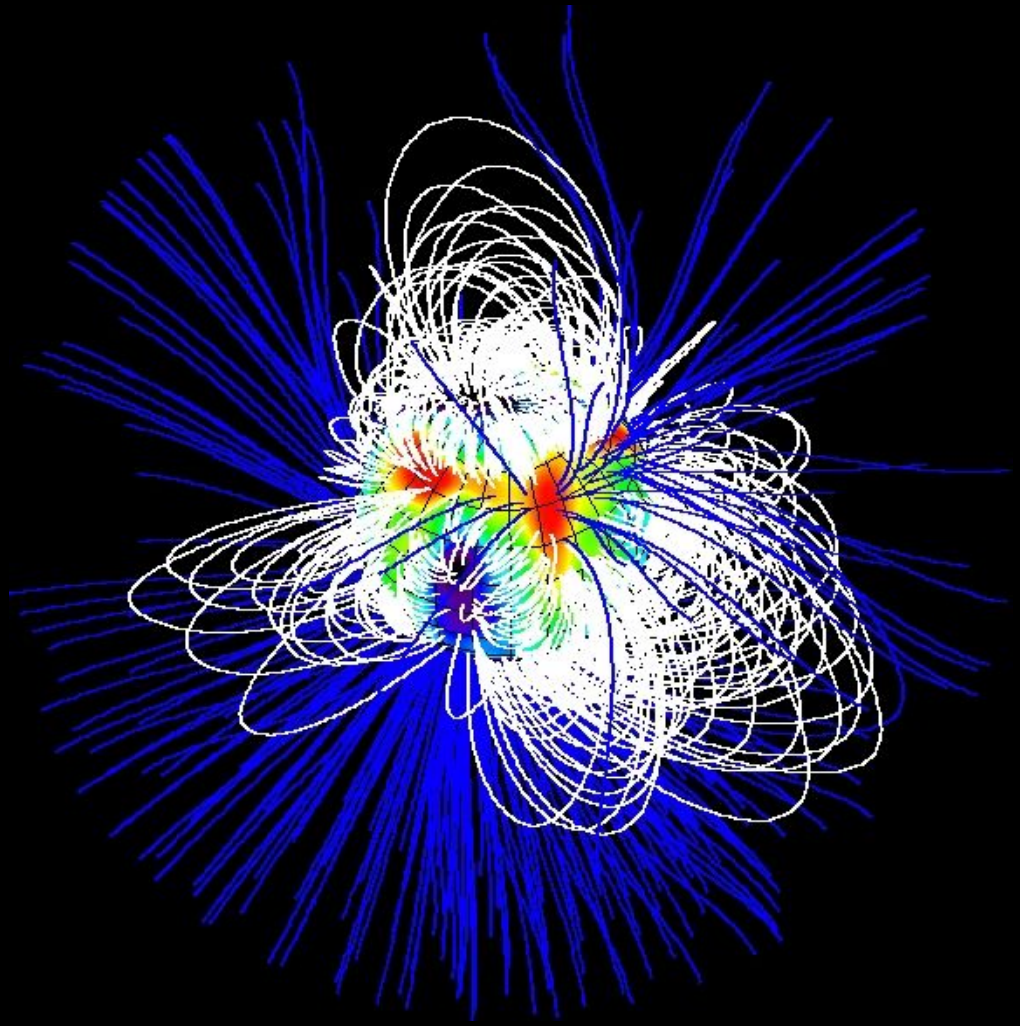


Image Credit: Pasal Petit

What is a massive star?

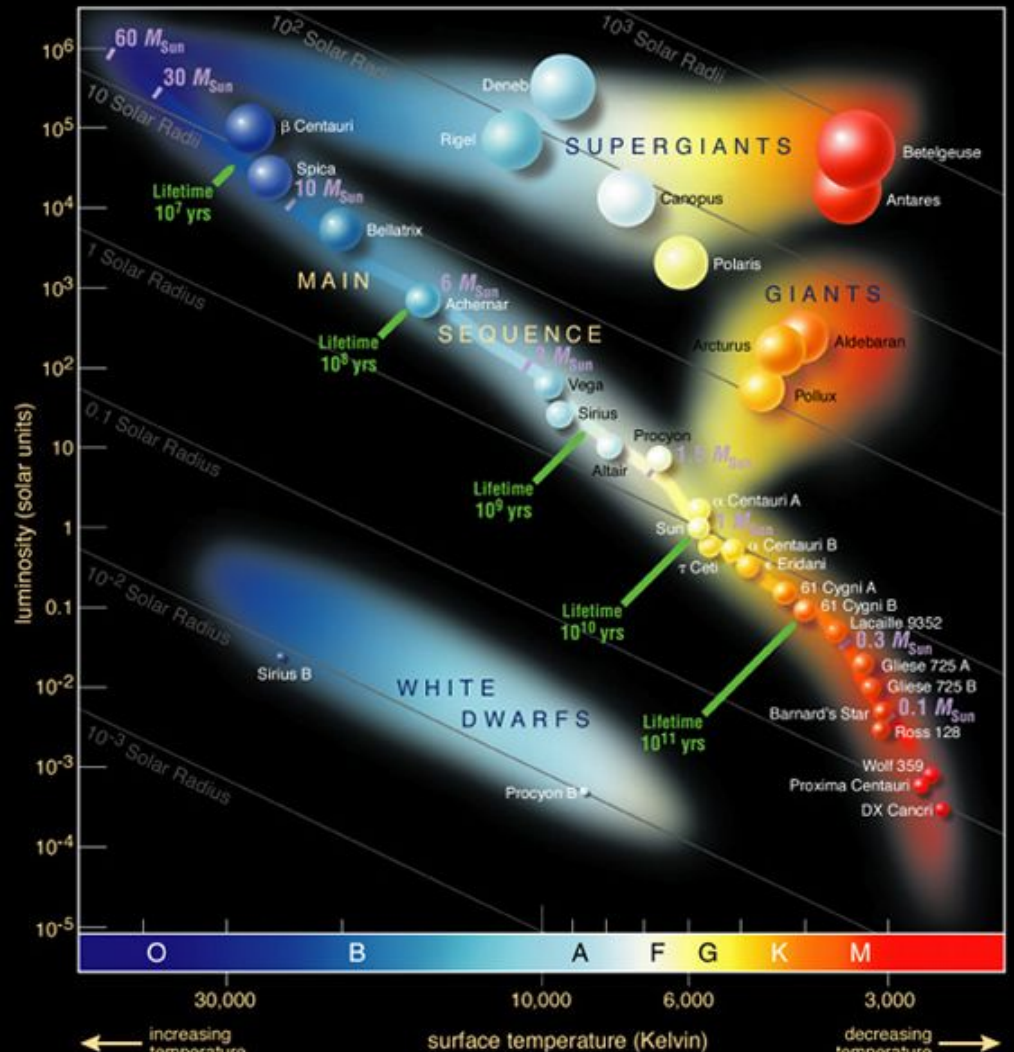


Image Credit: ESO

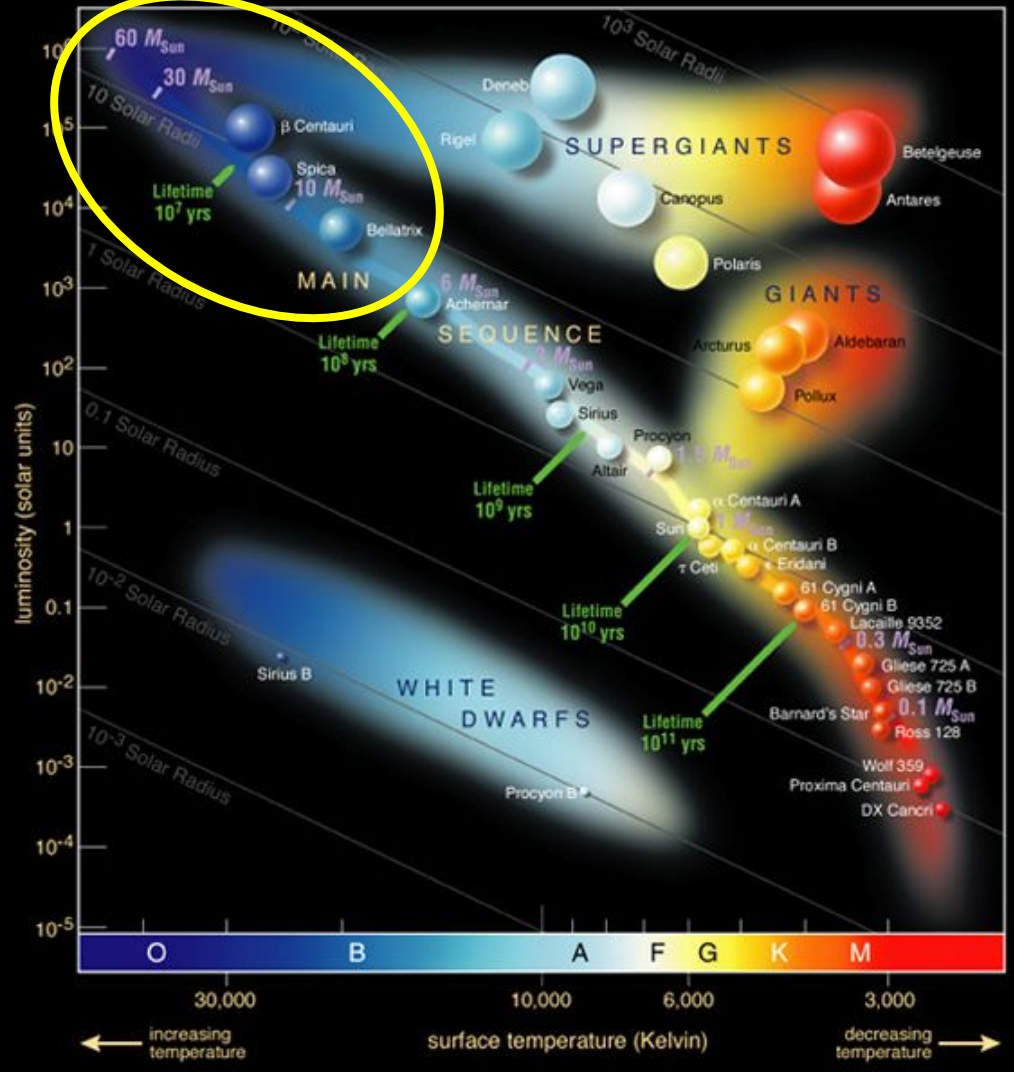


Image Credit: ESO

Massive stars are **heavy**.



200x

8x to 60x

1

yellow dwarf (Sun-like)

0.5x

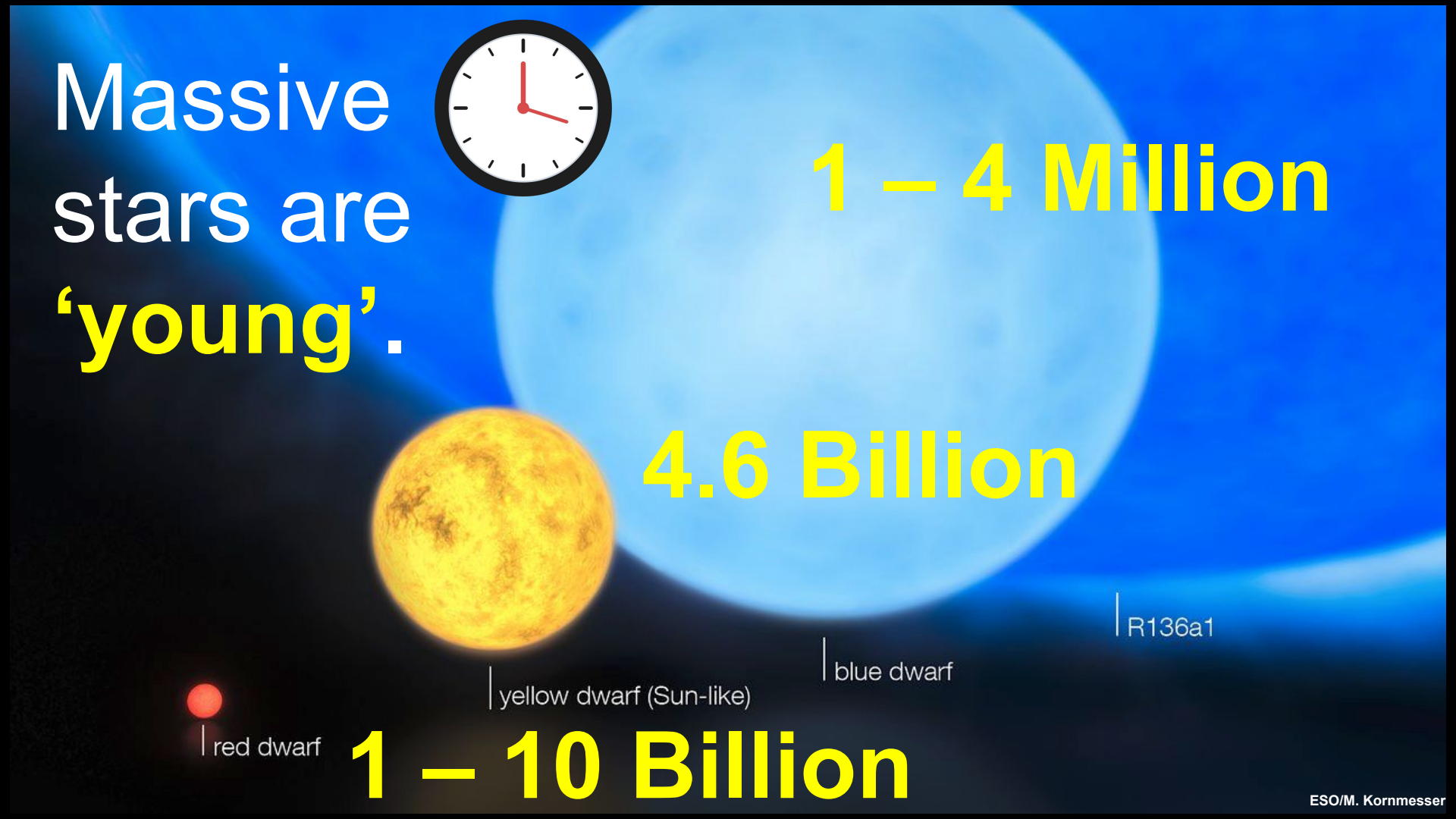
red dwarf

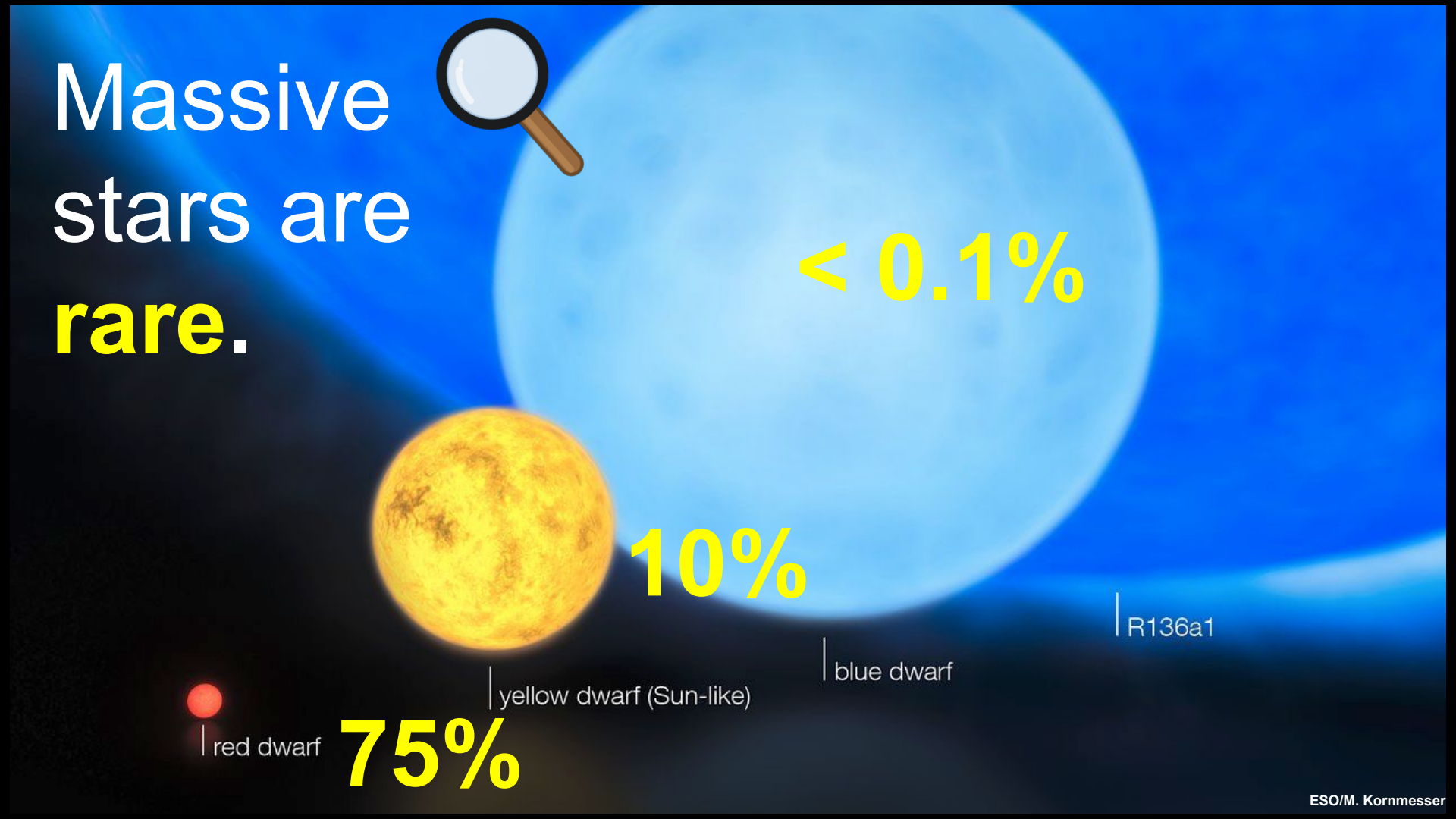
blue dwarf

R136a1

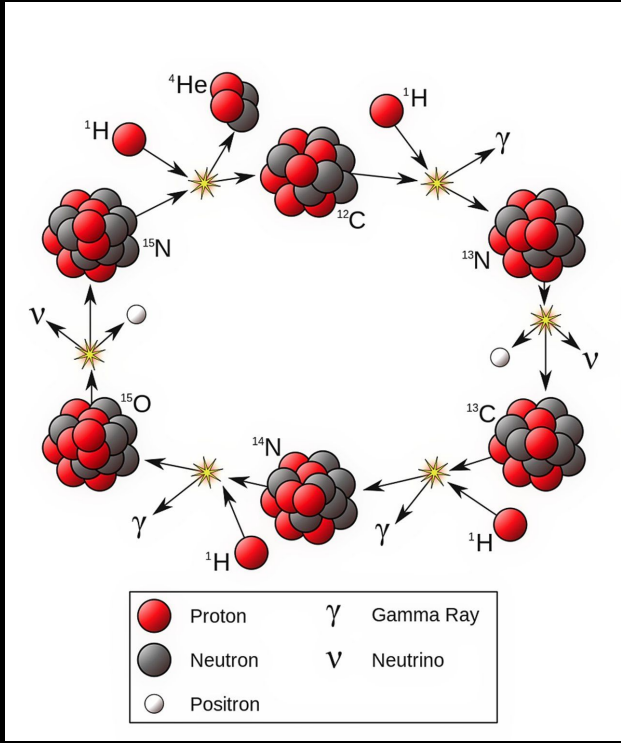
ESO/M. Kornmesser


Massive stars are born hot.
15,000 - 45,000°
6000°
blue dwarf
R136a1
3000°
yellow dwarf (Sun-like)
red dwarf
ESO/M. Kornmesser

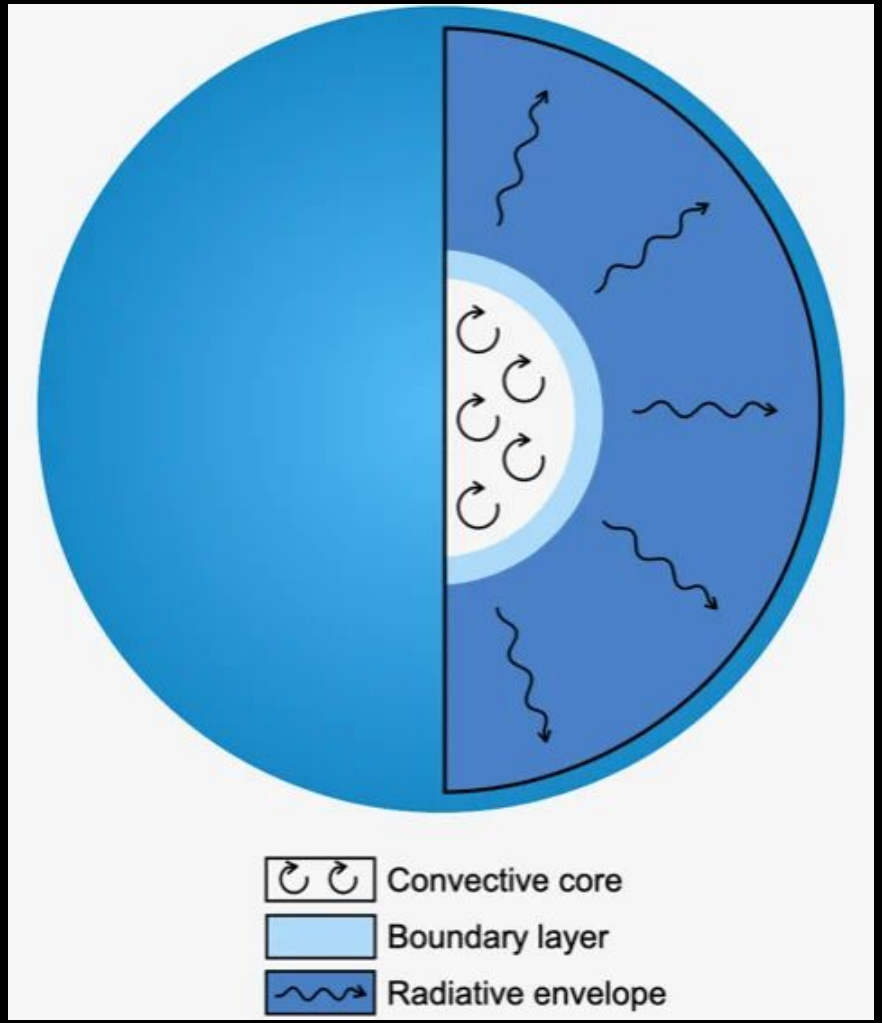

Massive stars are ‘young’.
1 – 4 Million
4.6 Billion
R136a1
blue dwarf
yellow dwarf (Sun-like)
red dwarf
1 – 10 Billion
ESO/M. Kornmesser


Massive stars are rare.
< 0.1%
10%
75%
R136a1
blue dwarf
yellow dwarf (Sun-like)
red dwarf
ESO/M. Kornmesser

4H→He + energy



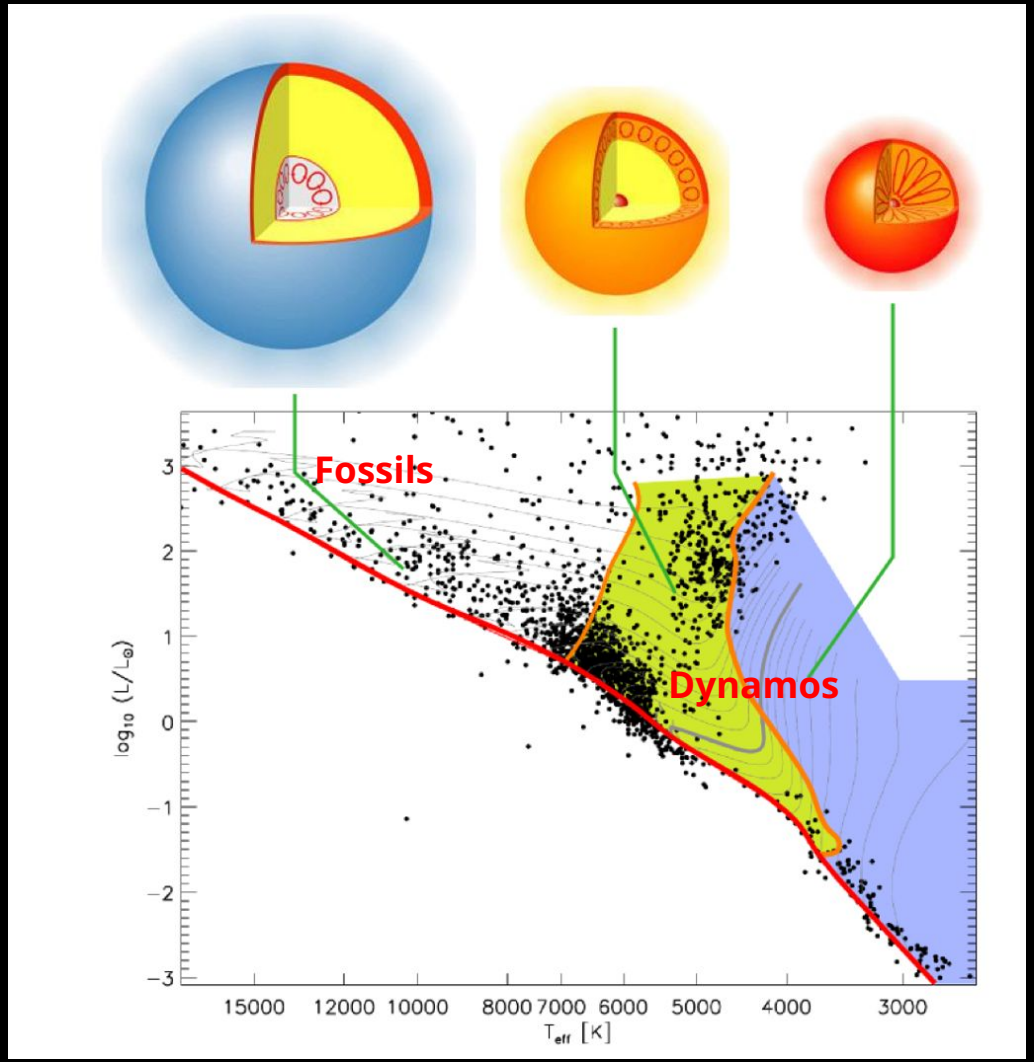
Credit: Borb



Credit: May Gade Pedersen

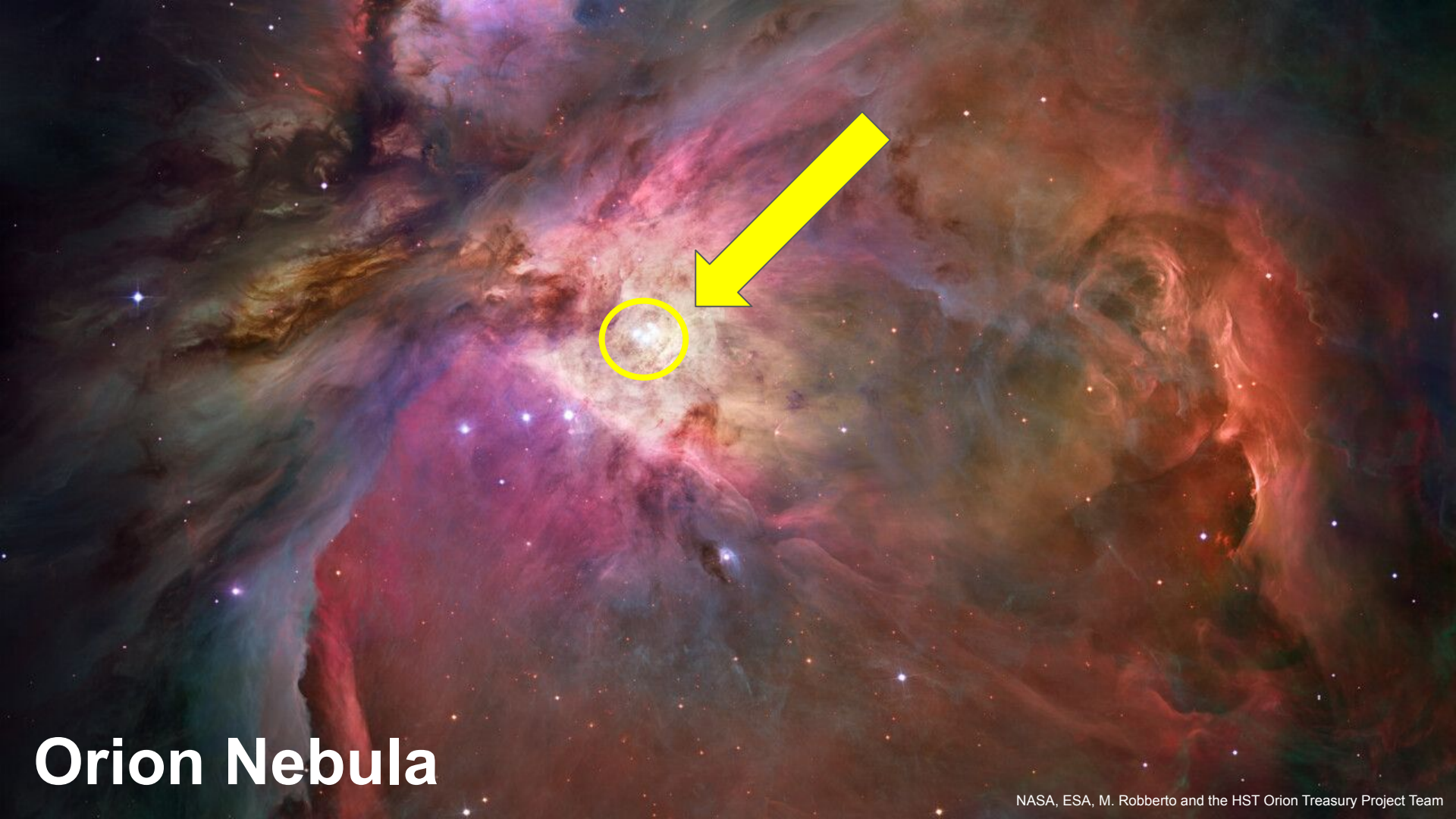
Magnetic Field Properties

- **Low mass:** common, complex, changing.
- **High-mass:** rare, simple, stable.



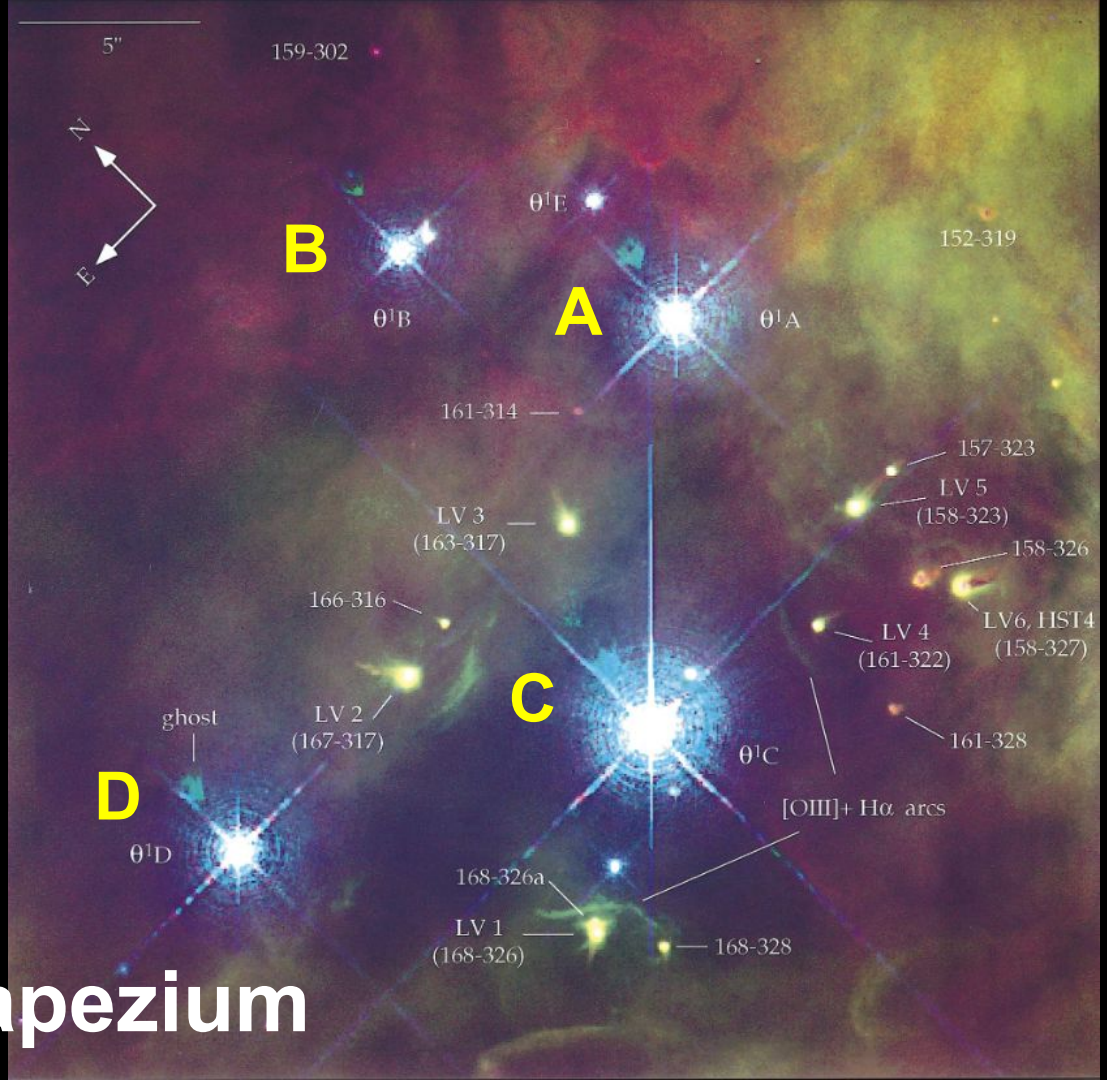


NASA, ESA, M. Robberto and the HST Orion Treasury Project Team



Orion Nebula

NASA, ESA, M. Robberto and the HST Orion Treasury Project Team

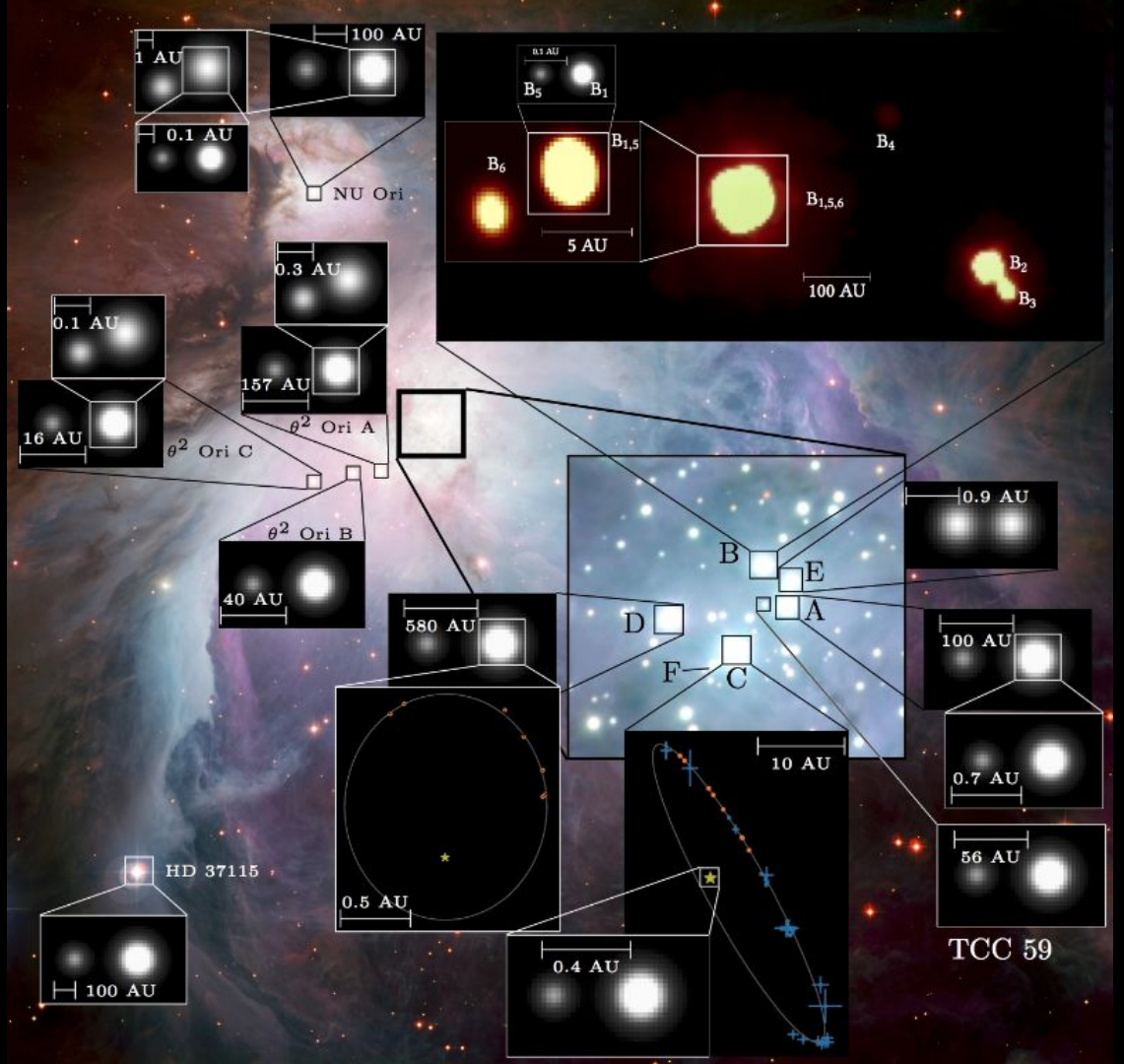


Orion Trapezium

Bally+ 1998

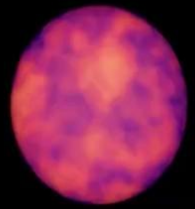
Massive stars are born as twins!

(or triplets or quadruplets or quintuplets…)

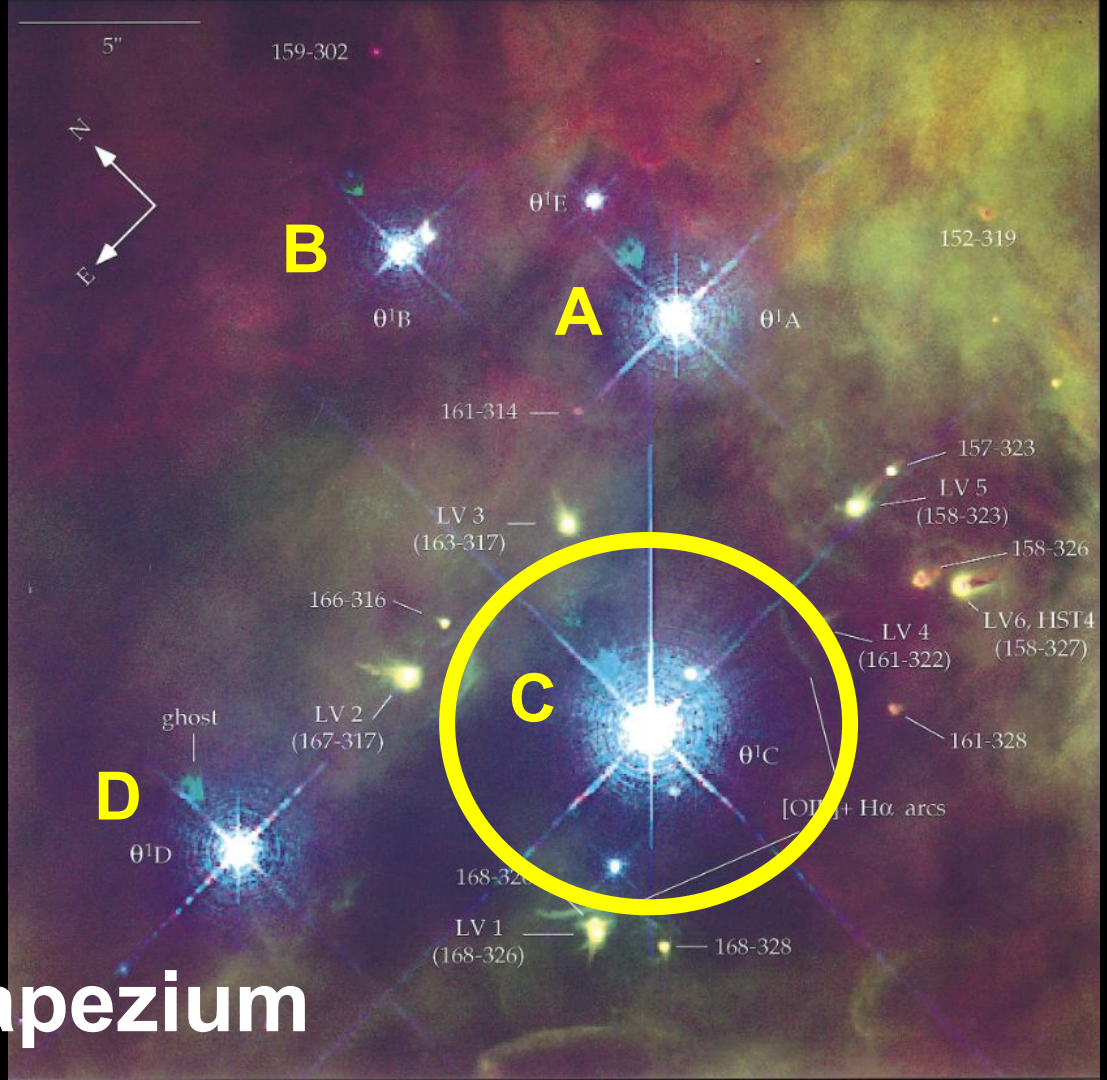


GRAVITY Collaboration+ 2018

Star Formation Simulation



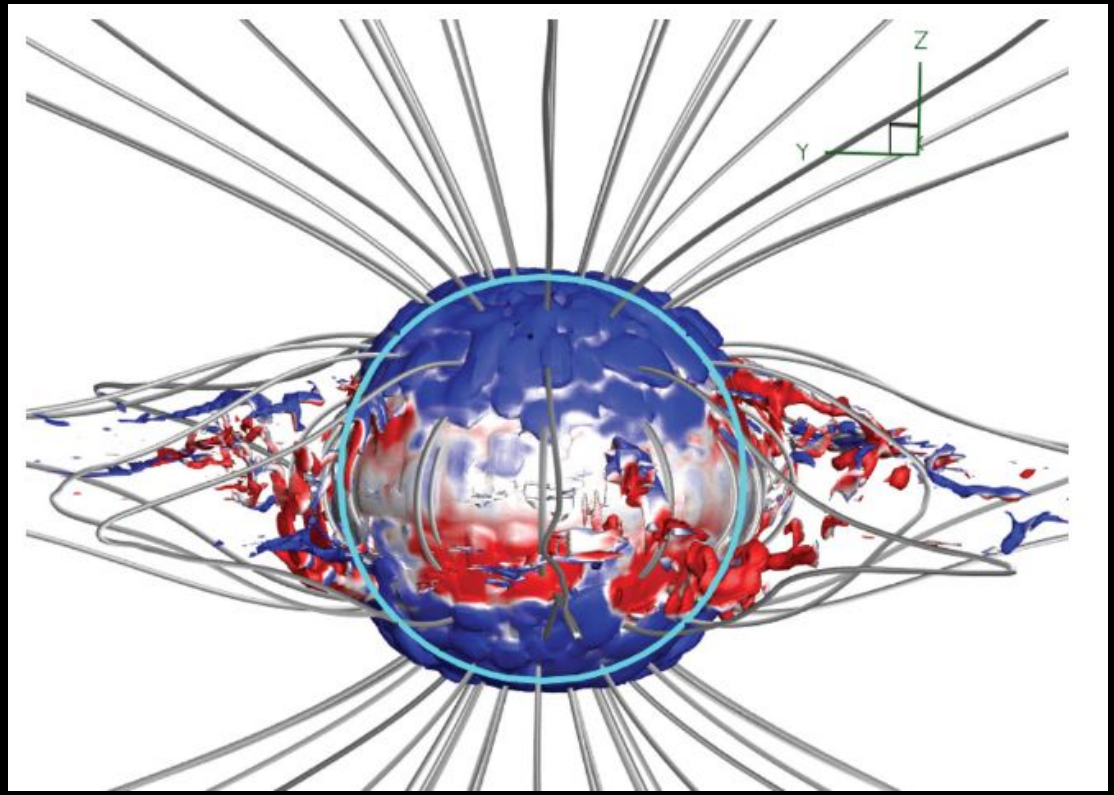
STARFORGE / Planetarium Mannheim



Orion Trapezium

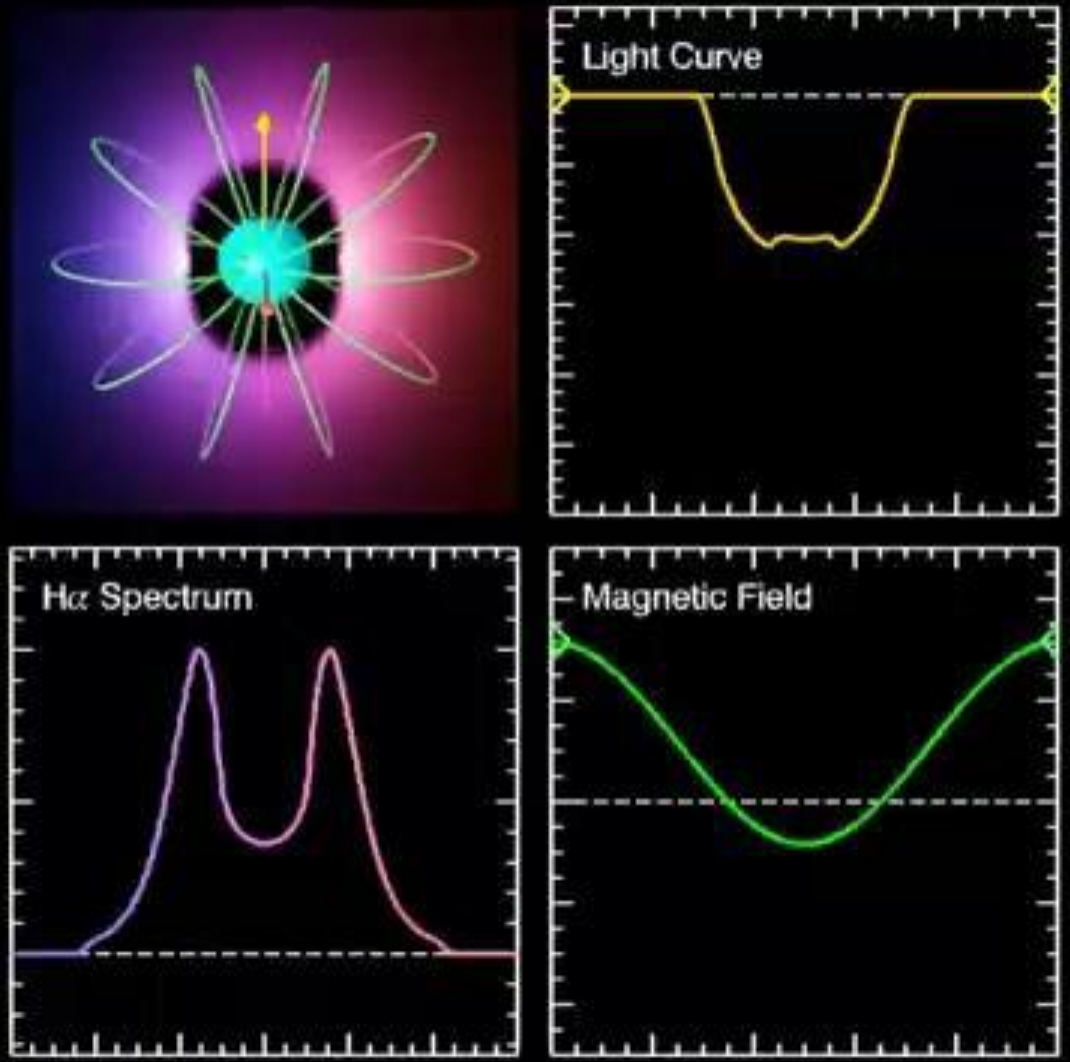
Bally+ 1998

- Confirmed magnetic in 2002.
- 1000x stronger magnetic field than the sun.



Θ^1 Ori C

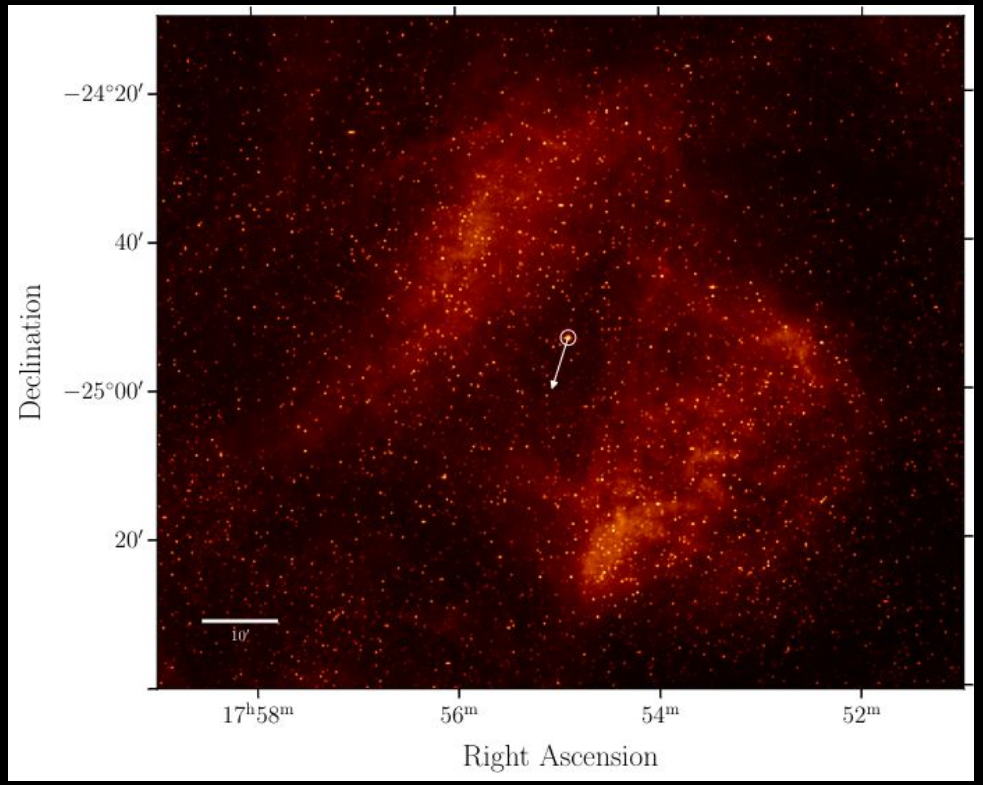
ud-Doula+ 2013

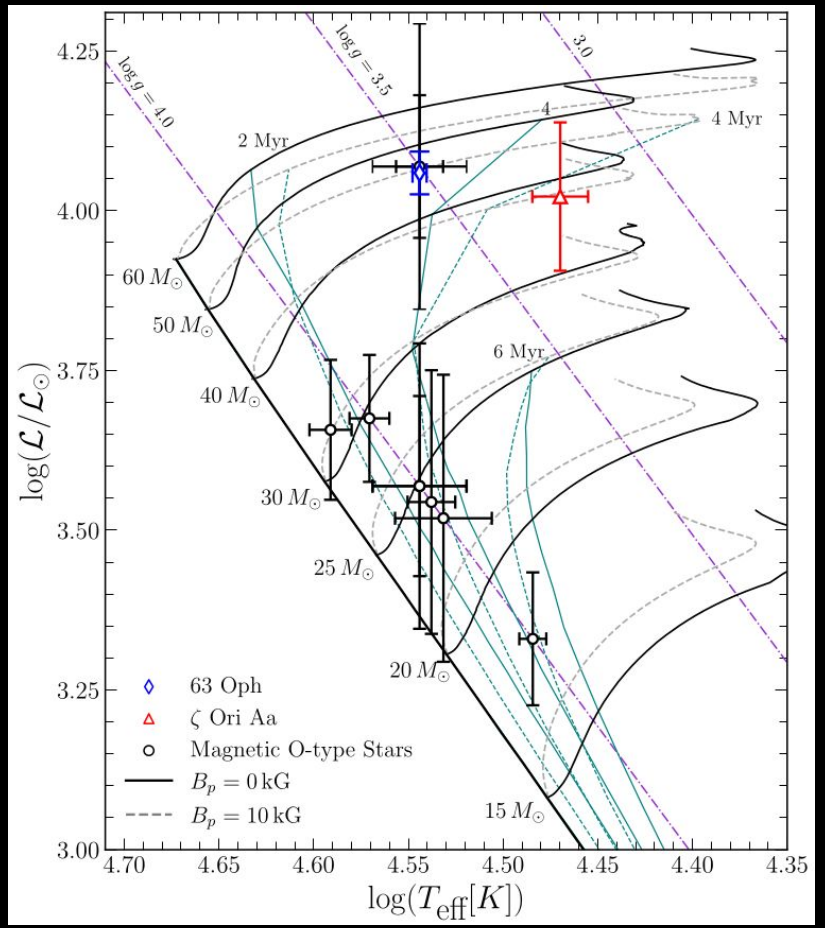


Credit: Rich Townsend

63 Oph

- **Only 11 confirmed magnetic O-type stars.**

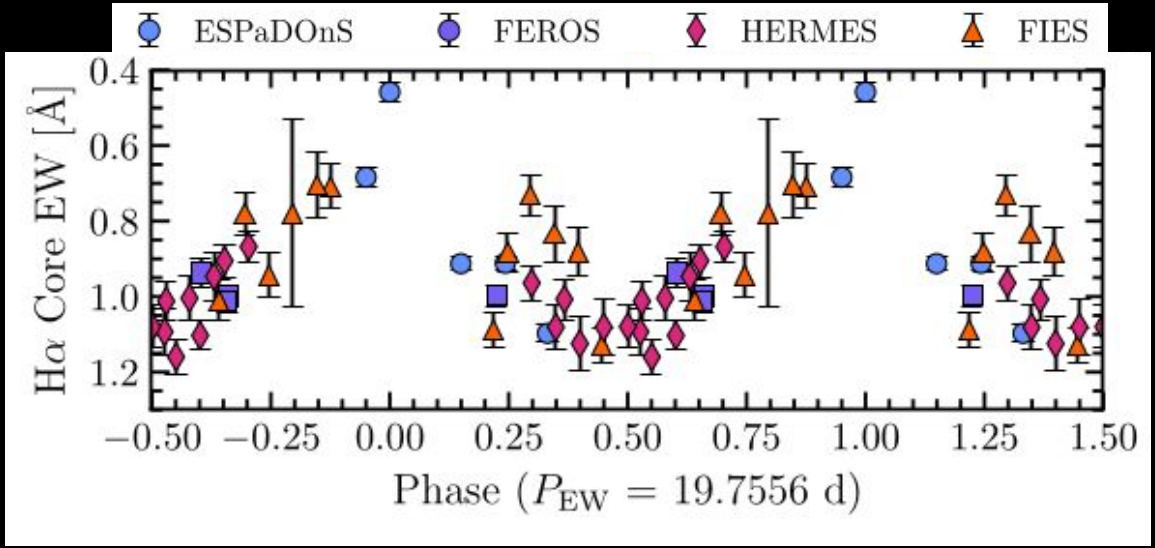


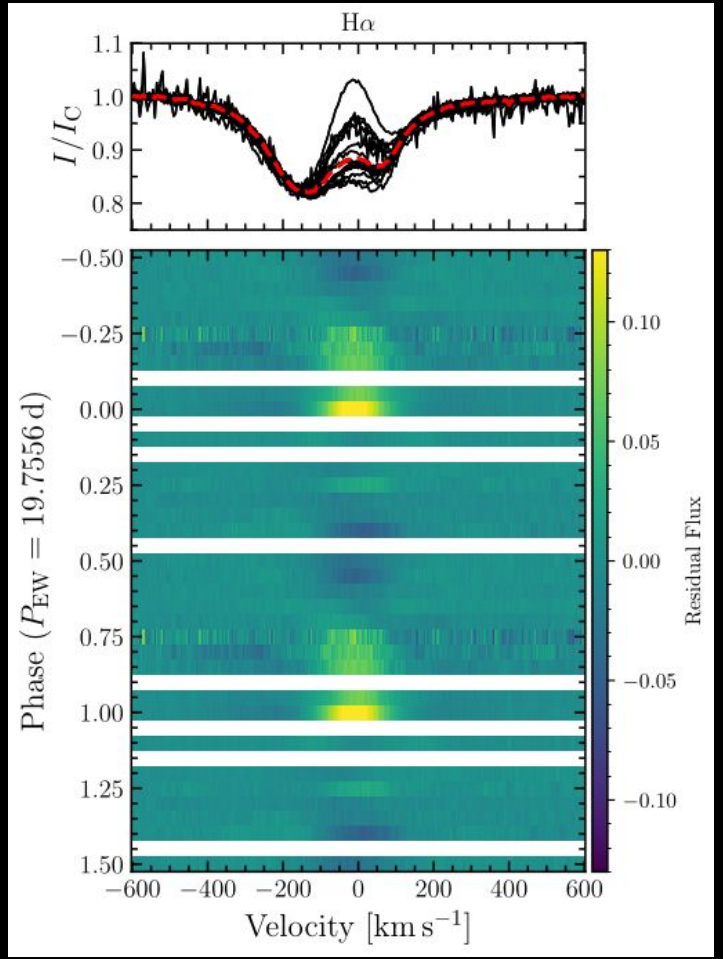


Barron+ 2025

63 Oph

- **Approximately 20 day rotation period**



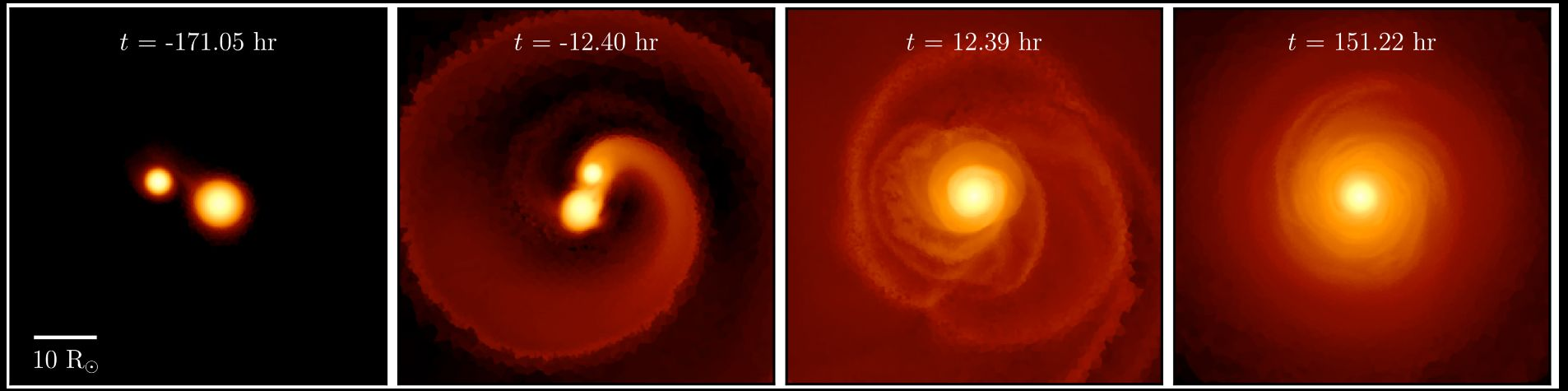


Barron+ 2025

How do you make a magnetic massive star?

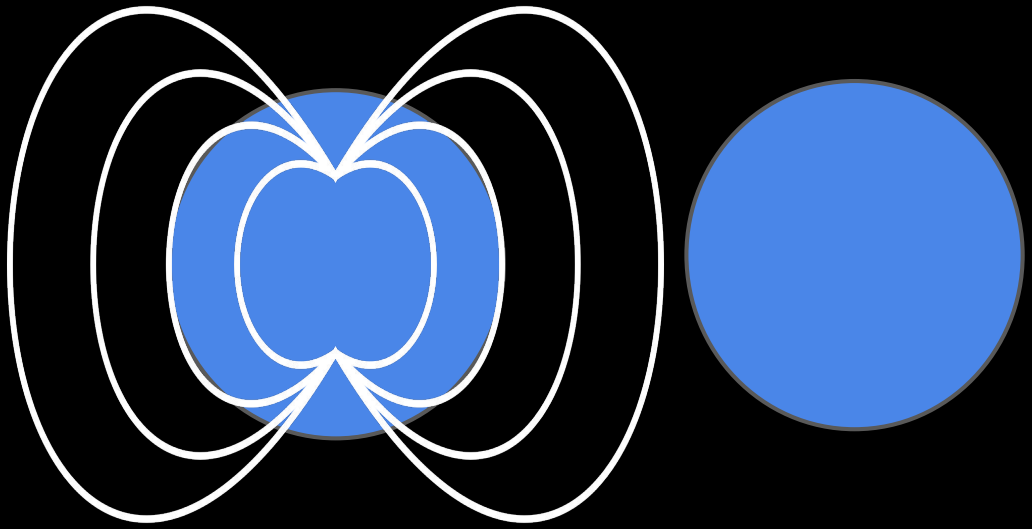
How do you make a magnetic massive star?

Merge with a binary companion!



Vynatheya+ 2026

30 year Binary Orbit



3 Million

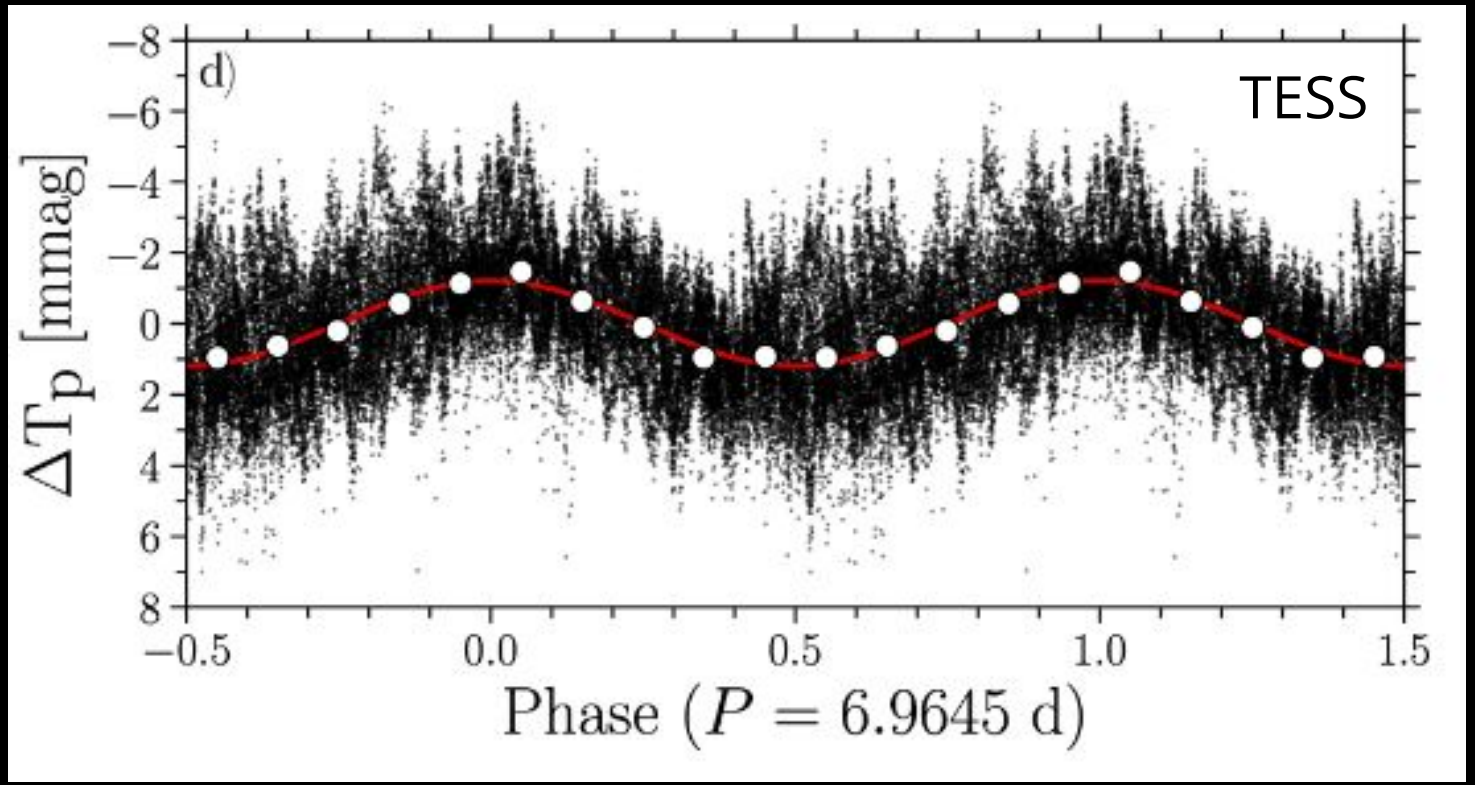
6 Million



HD 148937 & Dragon's Egg Nebula

Frost+ 2024, ESO/VPHAS+ team. Acknowledgement: CASU

HD 148937

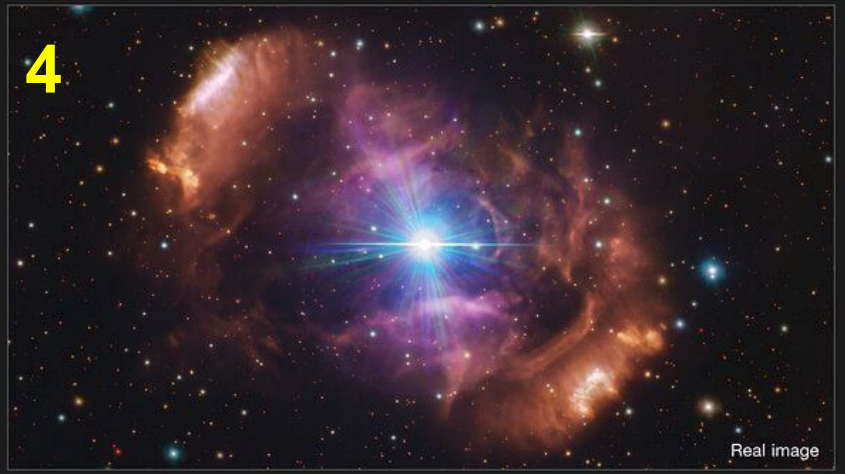


Frost+ 2024



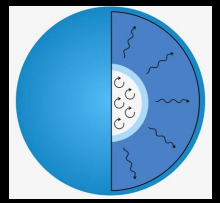




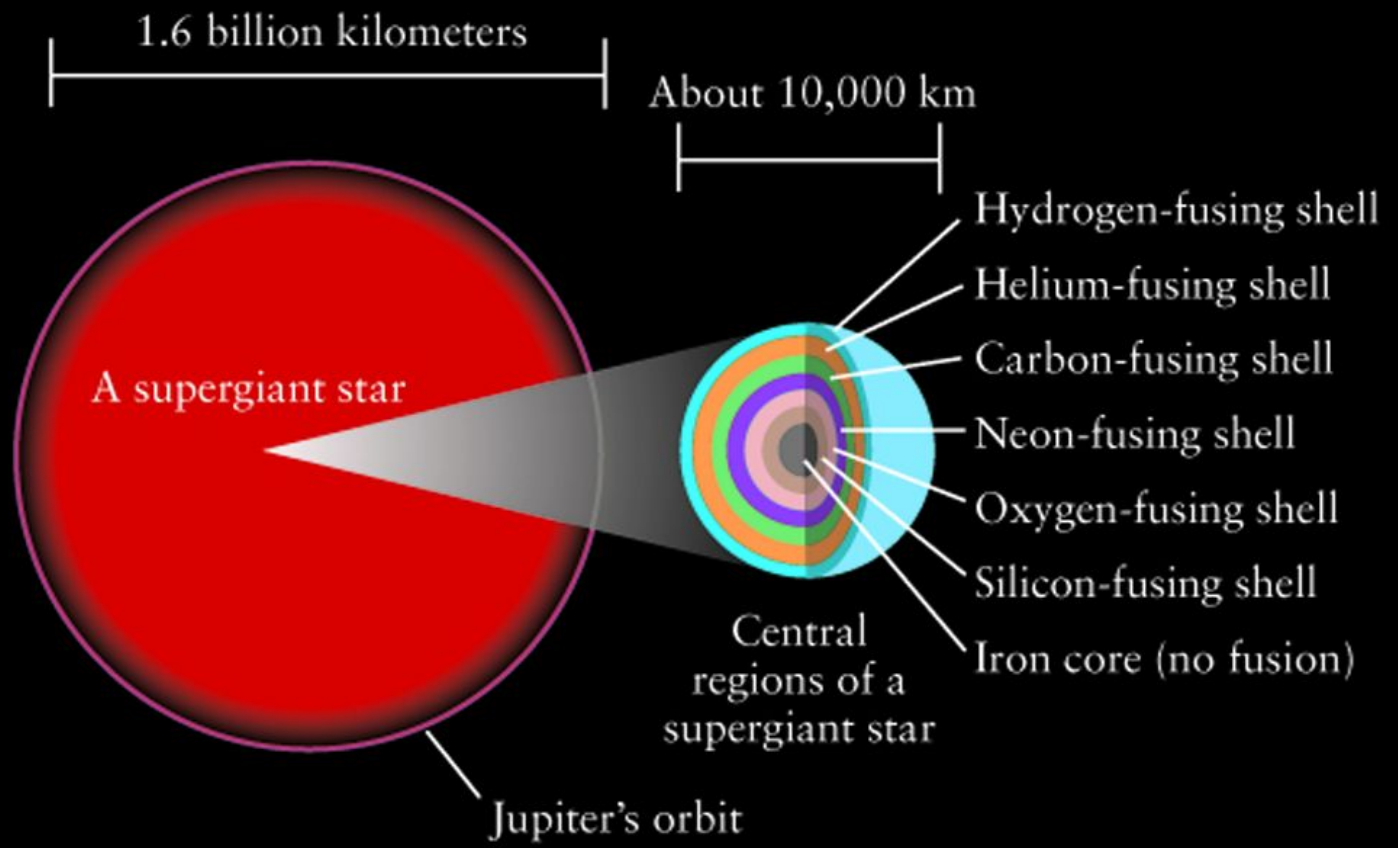


ESO/L. Calçada, VPHAS+ team. Acknowledgement: CASU

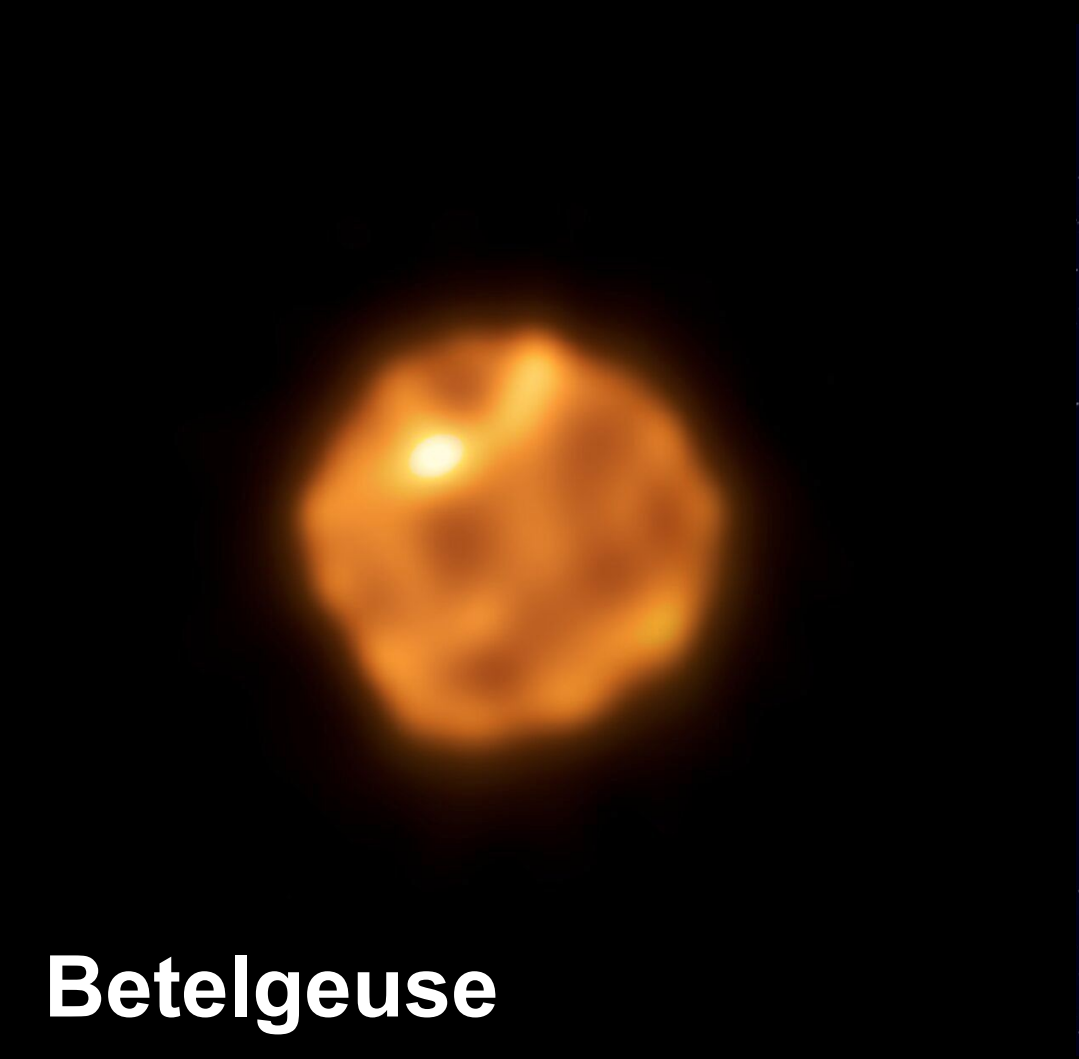
Late stage evolution



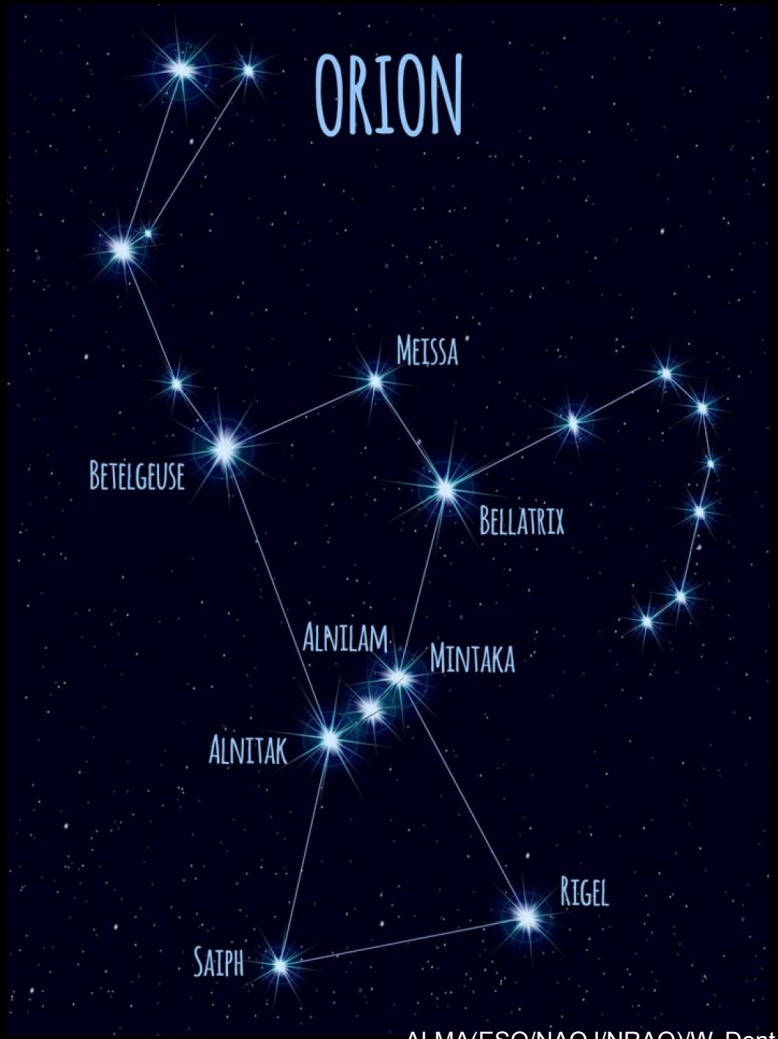
Core
H Depleted



What are the magnetic properties of evolved massive stars?



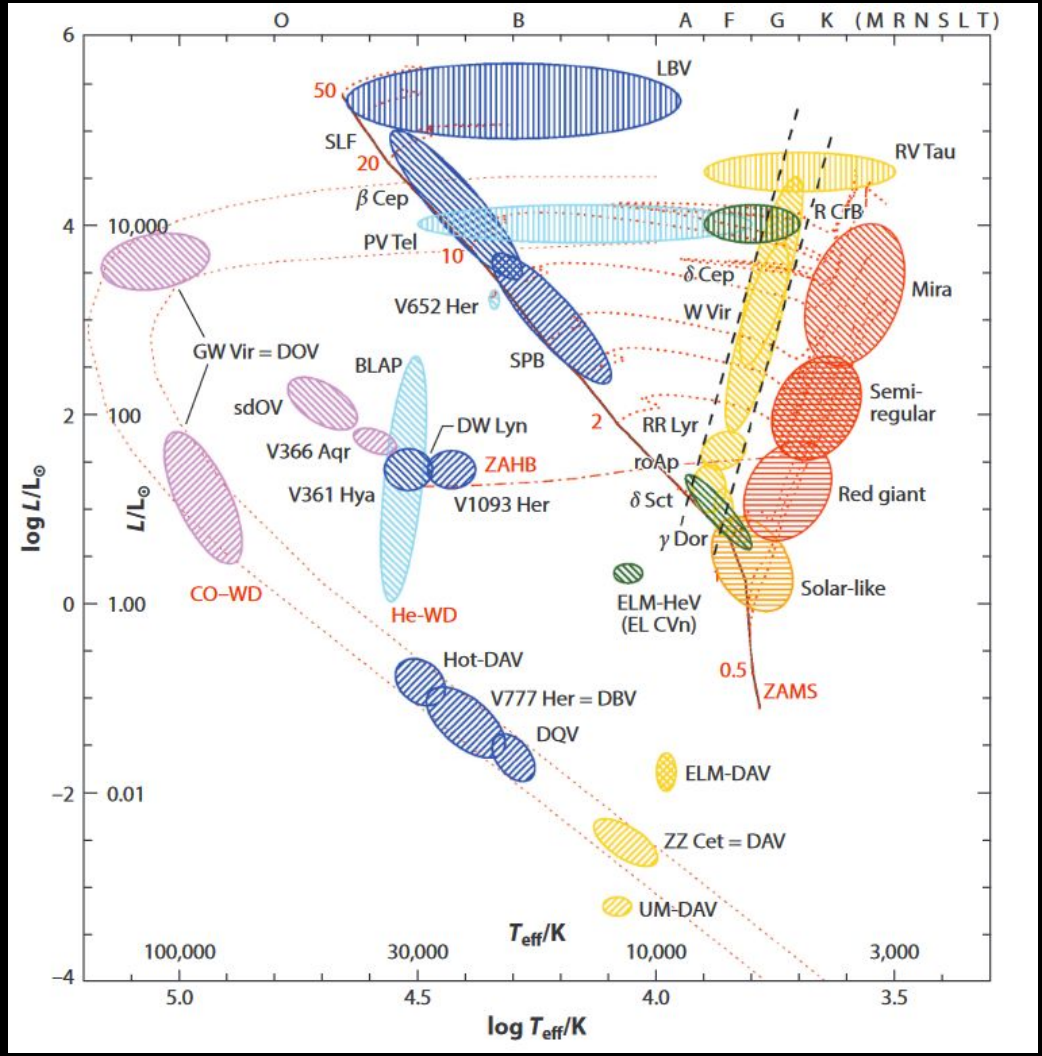
Betelgeuse



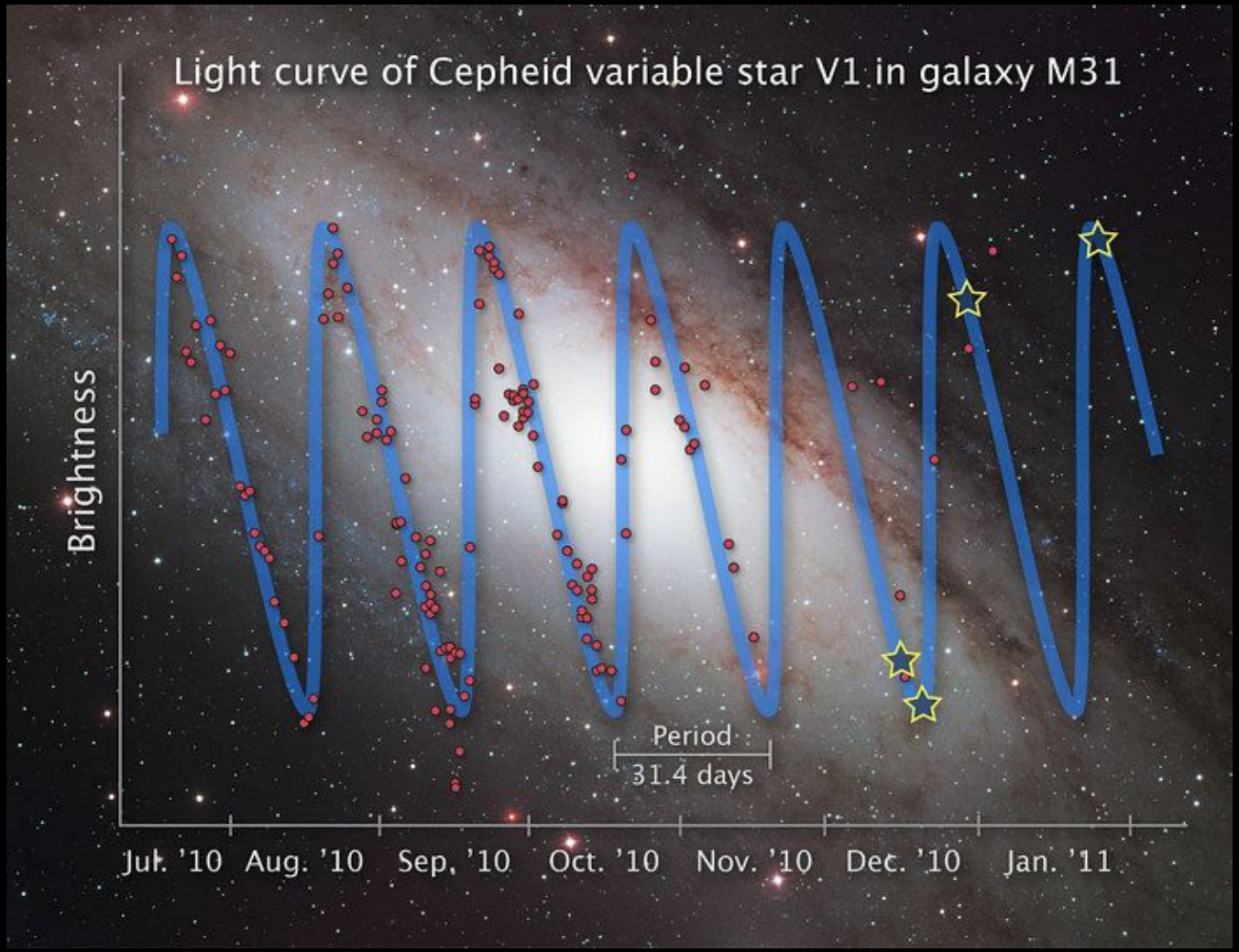
ALMA(ESO/NAOJ/NRAO)/W. Dent et al.

Betelgeuse Simulation

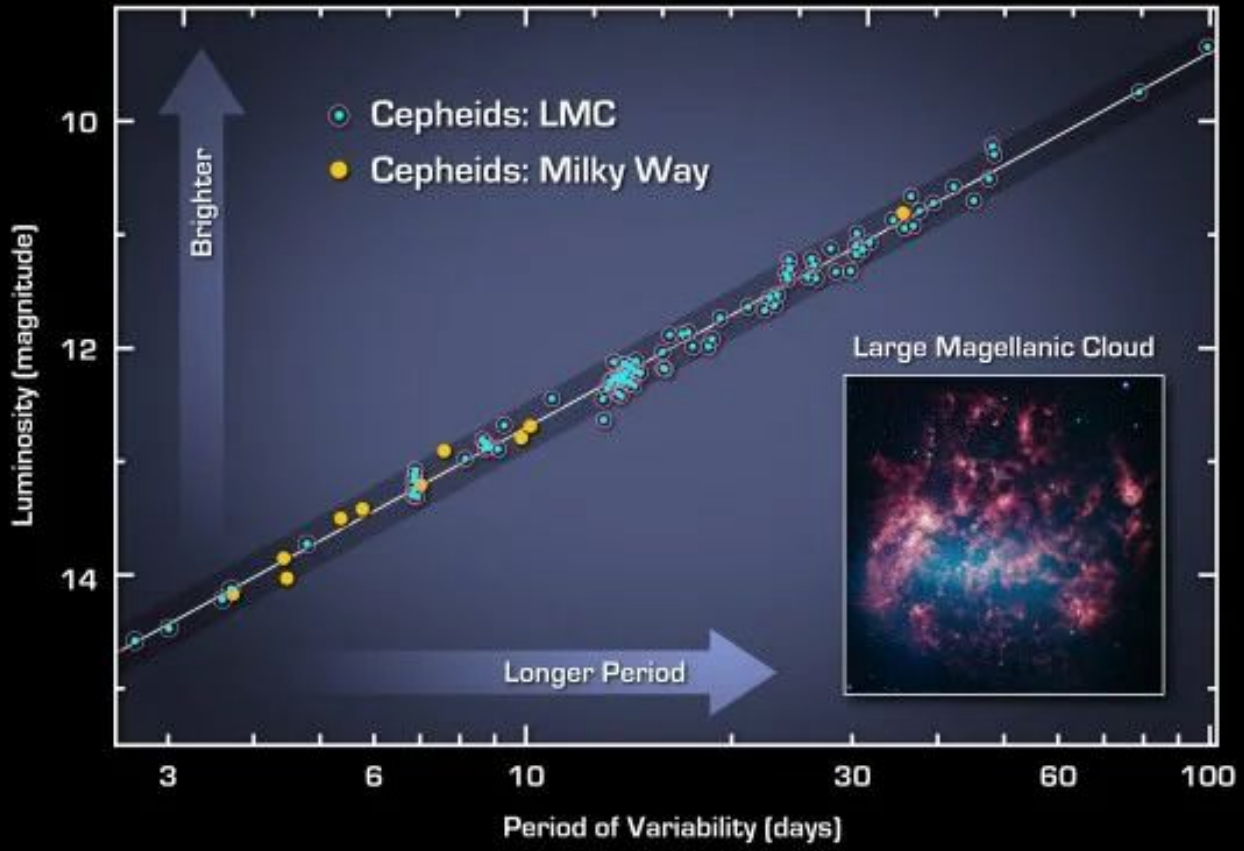
Classical Cepheids



Kurtz et al. 2022



NASA, ESA and Z. Levay (STScI). Science Credit: NASA, ESA, the Hubble Heritage Team (STScI/AURA) and the American Association of Variable Star Observers



Calibrated Period-luminosity Relationship for Cepheids

Spitzer Space Telescope • IRAC

NASA / JPL-Caltech / W. Freedman (Carnegie)

ssc2012-13a

Polaris the North Star

Distance

133 pc

System Components

Polaris (Aa + Ab) + B

Spectral Types

(F8Ib + F6V) + F3V

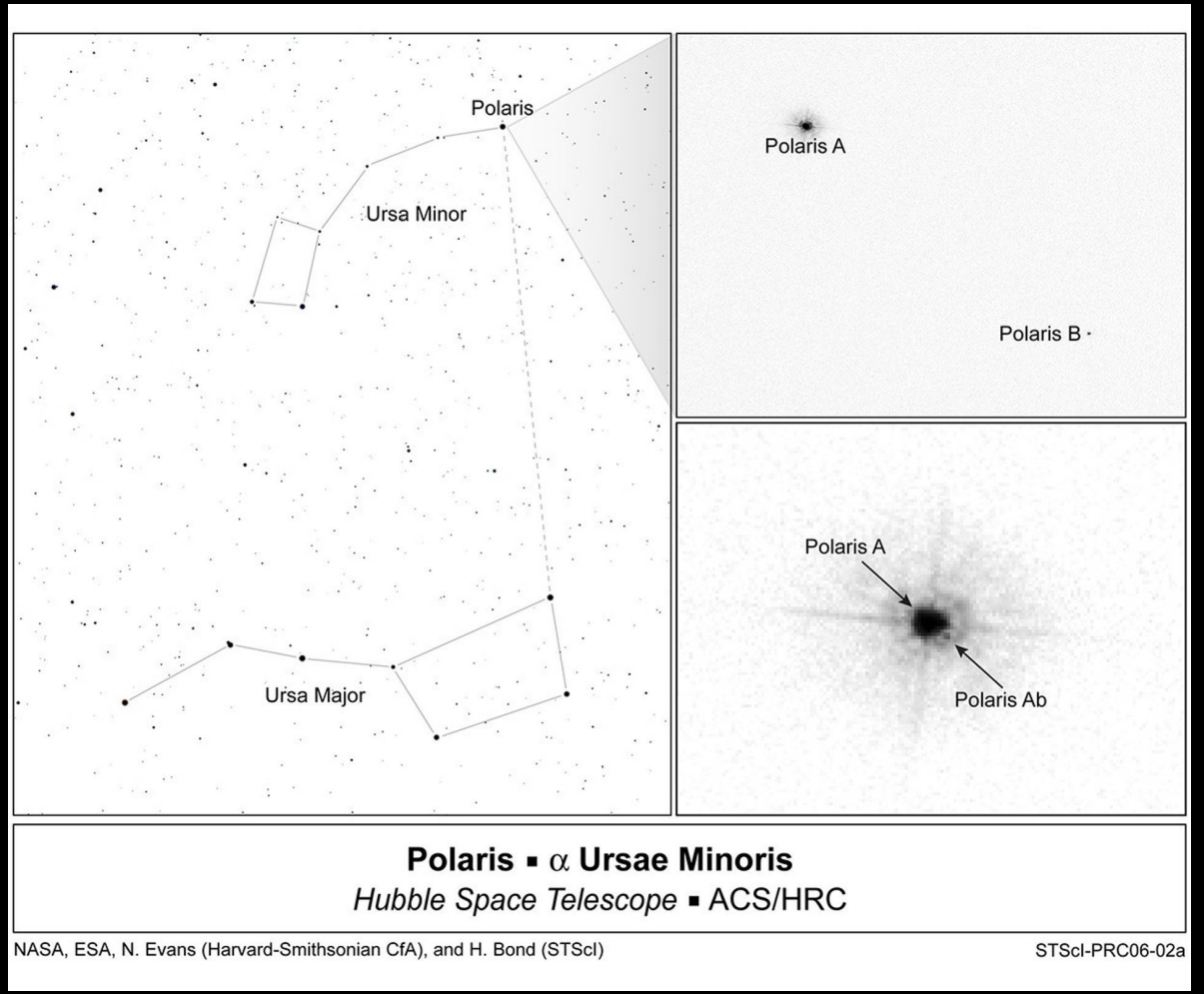
Inner Orbit (Evans+ 2024)

P_{orb} = 29.4 years

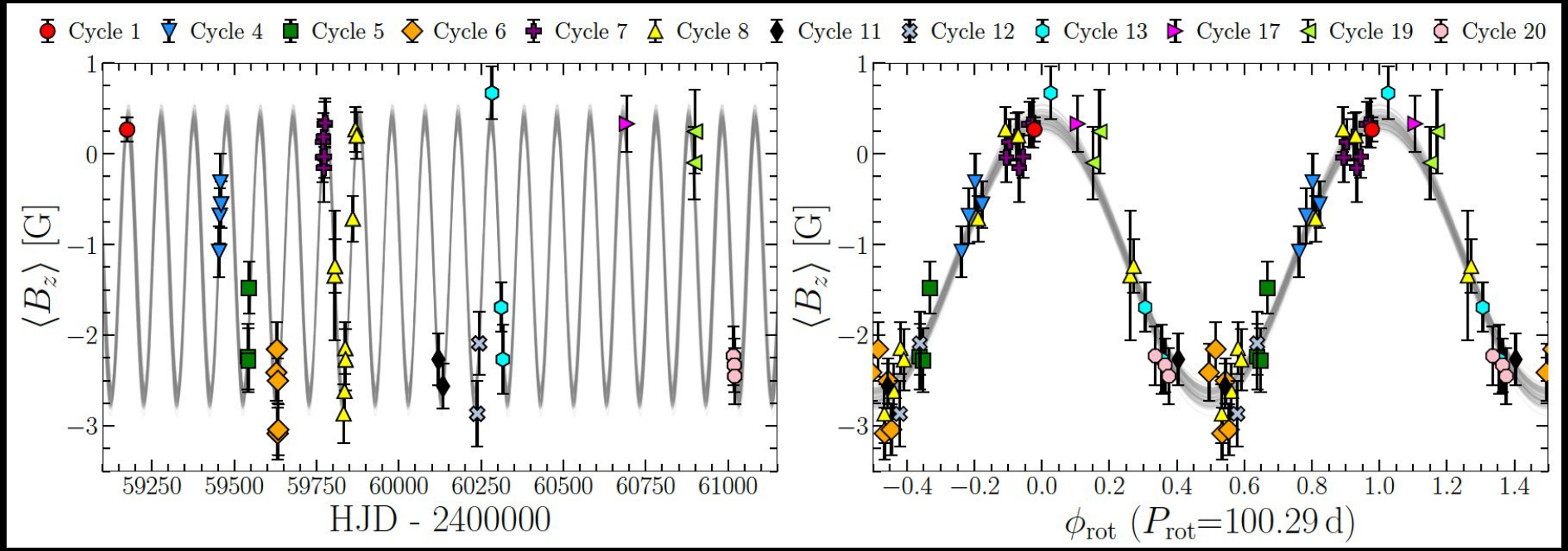
e = 0.62

Dynamical Mass

5.13 ± 0.28 $M_\odot$



First direct measurement of of the rotation period of a Cepheid



$$P_{\rm rot} = 100.29 \pm 0.19\,{\rm days}$$

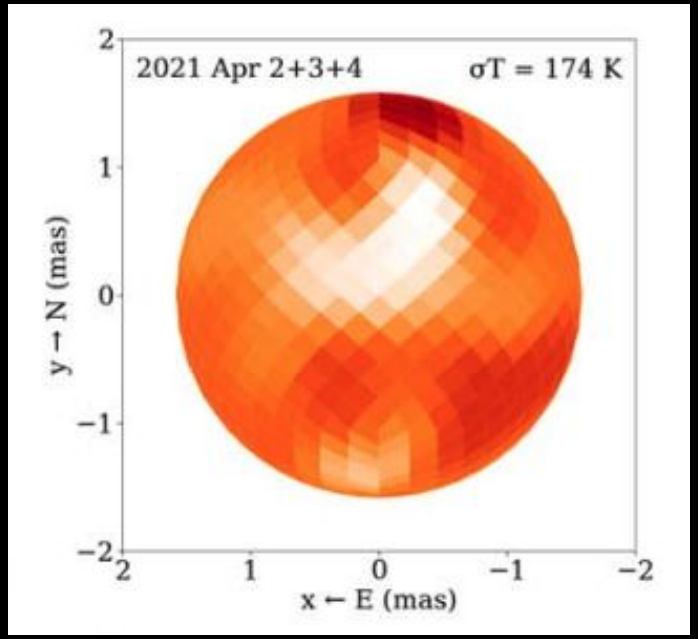
$$v_{\rm eq} = 23.3 \pm 0.2\,{\rm km\,s^{-1}}$$

$$v_{\rm eq} = \frac{2\pi\langle R_\star \rangle}{P_{\rm rot}}$$

Barron+ 2026

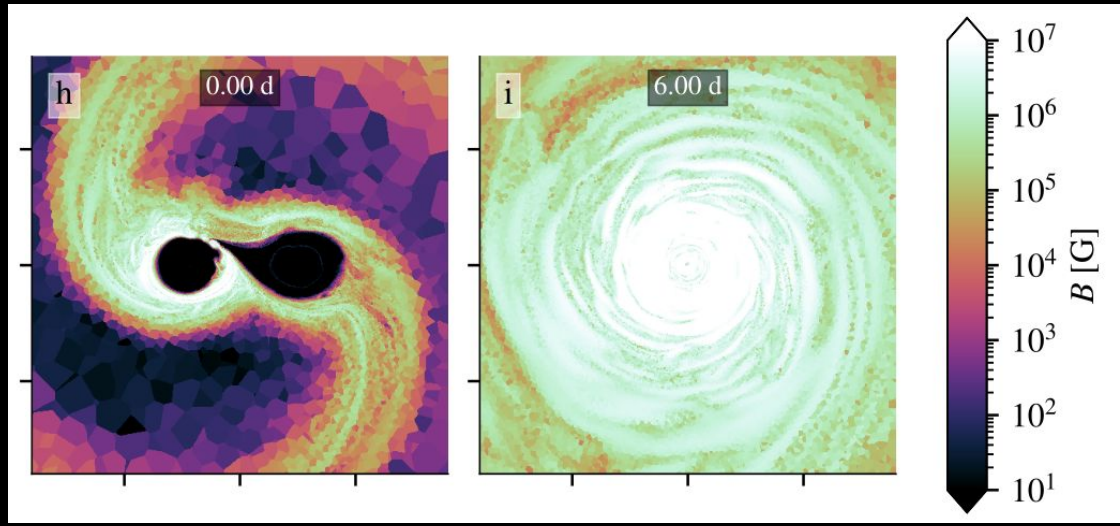
Uncertain origin of the magnetic field

Convective Dynamo?



Evans et al. 2024

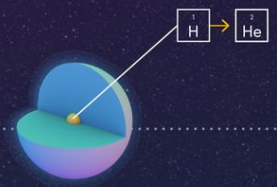
Fossil Field?



Schneider et al. 2019

MASSIVE STARS: ENGINES OF CREATION

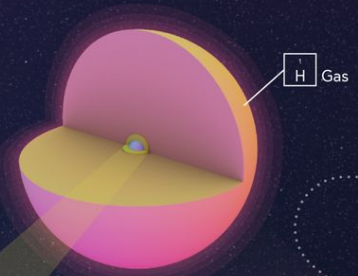
Through their cycle of formation, explosive "death" and reformation, massive stars (at least 8 times bigger than our sun) populate the universe with **new elements**, the raw material of all observable matter, including life.



1. MAIN SEQUENCE STAR

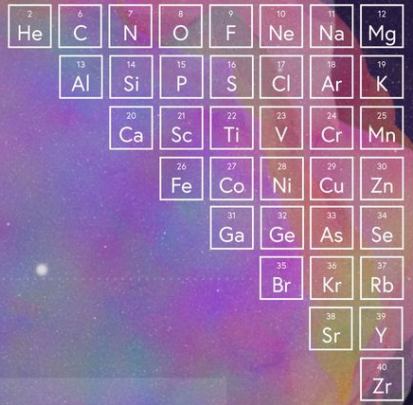
The star is fueled by hydrogen-to-helium fusion in its core.

Stage lasts 7 million years



2. DEVELOPMENT OF MULTI-SHELLED CORE

After hydrogen is exhausted in the core, other elements begin to fuse. Existing elements float above the denser center and form a multi-layered shell.

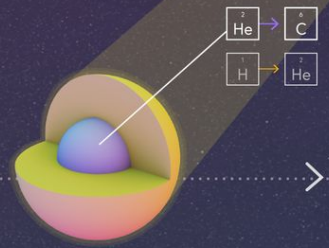


7. SUPERNOVA

The iron core cannot produce energy to balance the weight of the layered gas above. In less than one second, the core collapses and the upper layers rebound outward in a dramatic explosion, populating the universe with elements, the building blocks of creation.

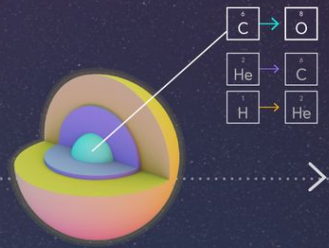
0.01 seconds

MULTI-SHELL CORE



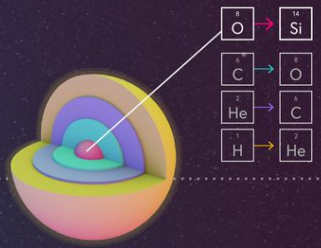
3. HELIUM-TO-CARBON FUSION

500,000 years



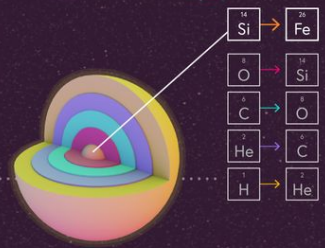
4. CARBON-TO-OXYGEN FUSION

600 years



5. OXYGEN-TO-SILICON FUSION

6 months

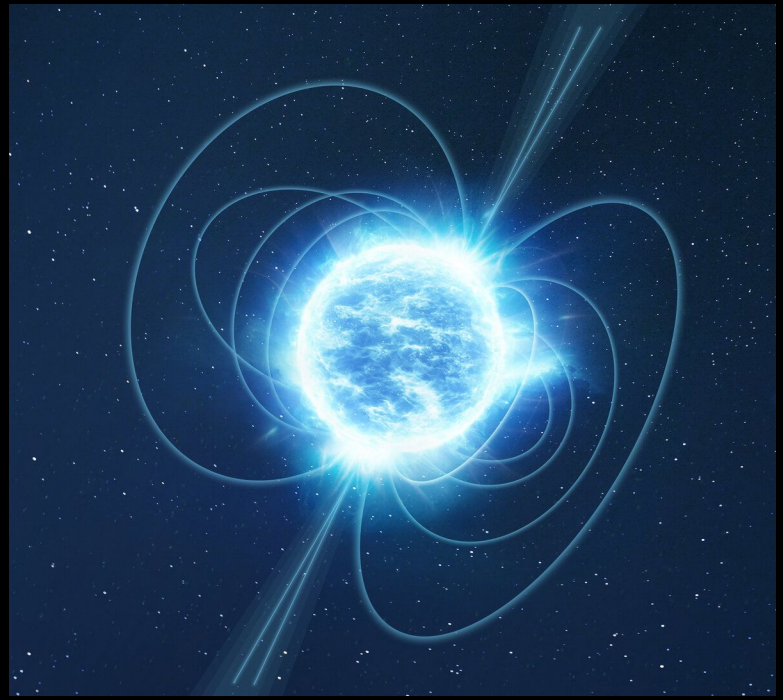


6. SILICON-TO-IRON FUSION

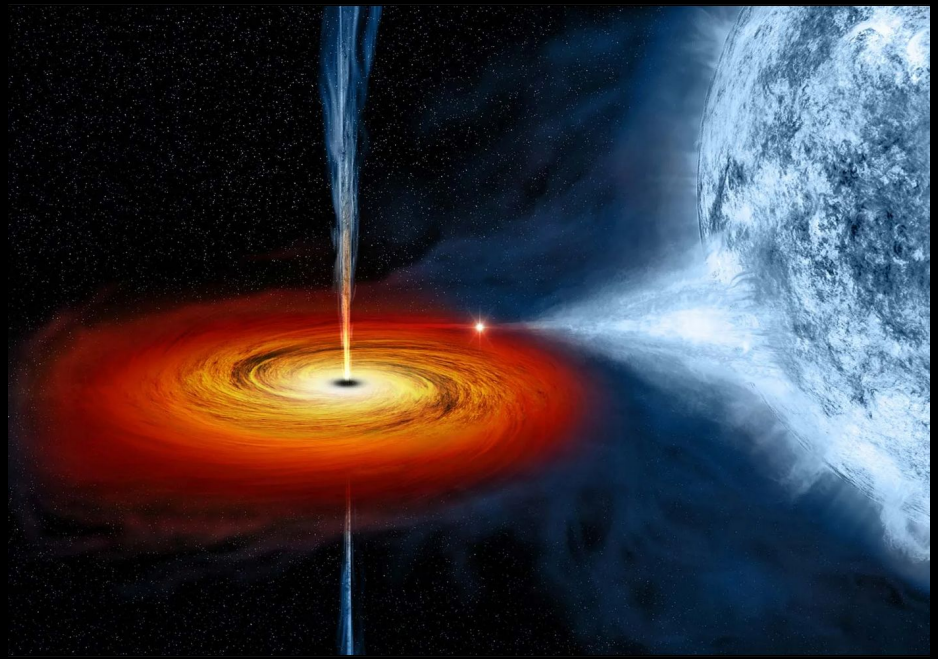
1 day

Image: NASA, ESA, Leah Hustak (STScI)

Compact Remnants

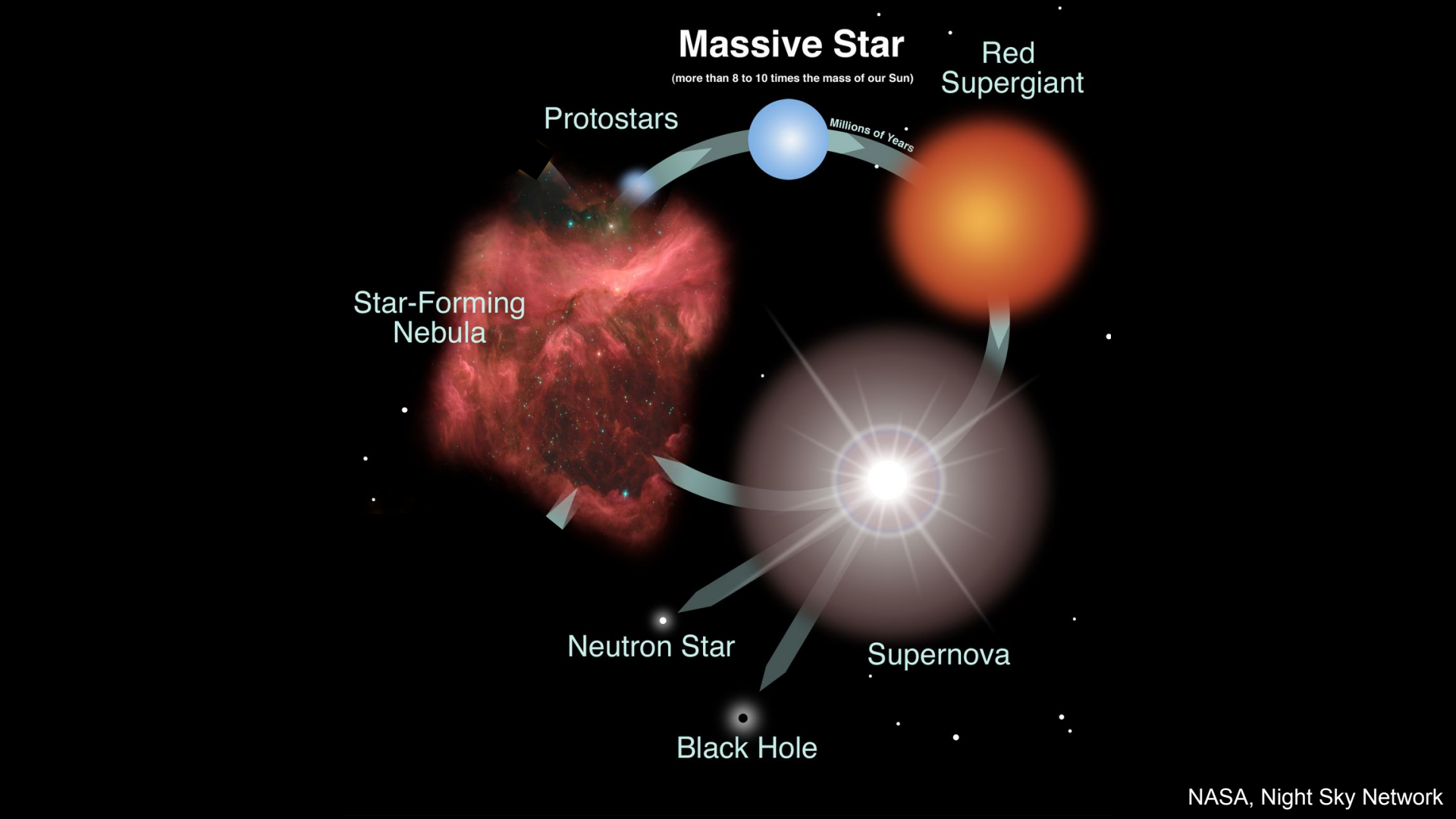


Neutron Star



Black Hole

CXC/M.Weiss/NASA



NASA, Night Sky Network